

Perceive, Externalize, Synthesize: Why Two-Pass Decomposition Beats End-to-End on Long-Form Multimodal Review

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Abstract

The default way to deploy a multimodal model on long-form review is end-to-end: feed it the raw audio or page images and ask for the review in one call. We show this default quietly *satisfices*: across 21 review cells, the end-to-end pass drops roughly a third of the source content and embellishes what it keeps—despite perceiving it, since 99.7% of the dropped content appears verbatim in the same model’s own transcript of the same source. A nine-experiment mechanistic study shows the bottleneck is neither perception nor the reasoning substrate—identical content is read equally well from text and from an *image* of that text—but **generation under load**: one pass cannot perceive, reason, and write a long faithful review at once. The mechanism dictates the fix: two same-weights passes—transcribe, then review the transcript—each with its own full output budget. This decomposition beats end-to-end on faithfulness on 18/21 cells and coverage on 16/21 (both sign-test significant), the gain scaling inversely with the end-to-end baseline ($r = -0.45$): the more the monolithic pass satisfies, the more decomposition recovers. The same headroom law organizes two further vendors and mixed-model pipelines, and predicts both observed failure modes. The win is the decomposition, not the modality.

Code: <https://github.com/19PINE-AI/cascade-gap> · Website: <https://01.me/research/cascade-gap>

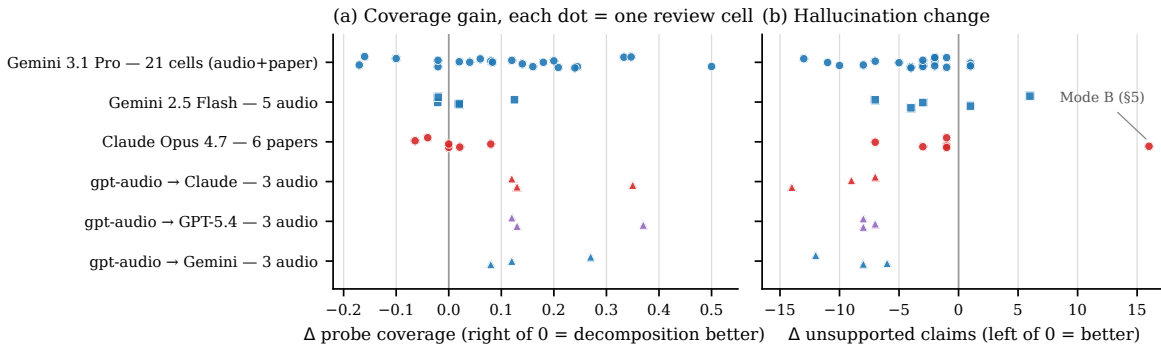


Figure 1. Results overview: decomposing long-form review into transcribe-then-review beats the end-to-end call, across models and pipelines. Each dot is one review cell: the change in (a) probe coverage and (b) unsupported-claim count from end-to-end (C_0) to two-pass decomposition (C_1), same weights per row (the gpt-audio rows are mixed pipelines vs. the native audio model’s end-to-end pass). On the headline 21-cell suite, coverage improves on 16/21 and hallucinations fall on 18/21 (sign tests $p=0.027$, $p=0.0015$); per-cell detail in Figure 8, and the regressions are the low-headroom and failure-mode cells of §5.

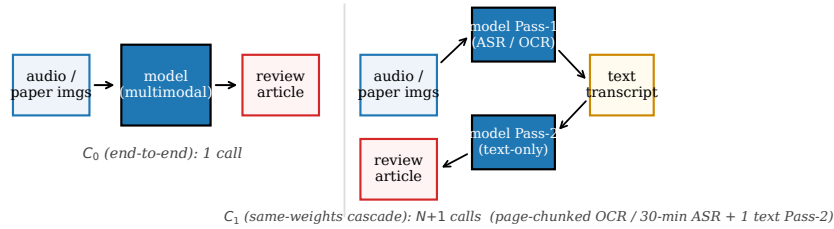


Figure 2. The two configurations, same weights. C_0 (end-to-end) reads the raw audio/page images and writes the review in one call. C_1 (two-pass decomposition) first transcribes the source (Pass-1, same weights and prompt as the reference pipeline), then writes the review from that transcript alone (Pass-2, text-only); only the reasoning step’s input representation differs.

1 Introduction

Multimodal foundation models read audio and document images directly, so the intuitive way to deploy them on a long-form review task—summarize this lecture, review this scanned report—is end-to-end: one call, raw source in, review out (Figure 2, top). Keeping the original modality in context looks strictly safer. No transcription step can lose information, and the model sees everything.

This paper argues, and measures, that the intuition is backwards for long-form review, and explains why. The argument runs in three steps.

The observation (§2). When a frontier model reviews a long source end-to-end, it quietly satisfices. On our running example—Andrej Karpathy’s 42-minute *State of GPT* talk—Gemini 3.1 Pro’s end-to-end review covers barely six in ten of the atomic facts an independent judge extracts from the source, and a quarter of its claims are unsupported (Figure 3). The failure is not perception: asked to simply *transcribe* the same audio, the same weights produce a transcript containing every single fact the review dropped. Across all 21 cells, 99.7% of the content the end-to-end pass drops sits verbatim in the model’s own transcript (Figure 4). The model perceives the content and then loses it downstream.

The mechanism (§3). Where downstream? Three hypotheses fit the observation: text is a better reasoning substrate than raw audio/pixels (H1); a single pass cannot perceive, reason, and write a long faithful review at once (H2, generation under load); attention dilutes over long low-density modality tokens (H3). Nine experiments (E1–E9) separate them. The substrate hypothesis fails everywhere we can test it: identical propositional content is read equally well from text and from an *image* of that text—at the activation level on open VLMs, and behaviorally on the frontier models themselves, where under reasoning load the image is, if anything, *better* (Figure 5). What does bind is generation: collapsing transcription and review into one call crashes coverage or produces no review at all, while re-attaching the raw modality to a transcript-equipped pass changes little (Figure 6). The bottleneck is the single pass, not the pixels.

The solution (§4). The mechanism dictates the fix: give each step a dedicated, full-budget pass. Pass 1 transcribes the source (*perceive and externalize*); Pass 2 writes the review from the transcript alone (*synthesize*), same weights throughout. This two-pass decomposition (C_1) beats end-to-end (C_0) on faithfulness and coverage across the 21-cell suite (Figure 1), and the gain obeys an *inverse-baseline law*: it scales with how badly the monolithic pass satisfices ($r = -0.45$, Figure 7). The same law—not a uniform win—organizes the cross-vendor picture (Figure 9)

Running example: Gemini 3.1 Pro reviews Karpathy’s *State of GPT* (42-min audio), end-to-end. All excerpts are verbatim from the released run artifacts.

The talk says: “*The vocabulary size is usually a couple ten thousand tokens*” — and the model hears it: its own transcript of the audio contains the sentence verbatim.

The end-to-end review writes: “*A typical vocabulary size is around 10,000 tokens.*” — one of **13 unsupported claims**, alongside a misdescription of RLHF as a fourth serial pipeline stage.

The talk ends with Karpathy prompting GPT-4: “*can you say something to inspire the audience of Microsoft Build 2023?*” — again present verbatim in the model’s own transcript.

The end-to-end review: drops it entirely, one of **19 of 51 probes missed**. All 19 missed probes are present in the model’s own 8,500-word transcript of the same audio.

The two-pass review (§4: same weights, written from that transcript alone) covers **49 of 51 probes with 3 unsupported claims**, and gets the example right: “*Vocabulary sizes are usually a couple ten thousand tokens (e.g., 50,257).*”

Figure 3. The observation, on one real cell. The end-to-end pass satisfices—it drops content it demonstrably perceived and embellishes what it keeps—while the same weights, decomposed into transcribe-then-review, recover both faithfulness and coverage. Karpathy is the bolded row in Figure 8a.

and predicts exactly two failure modes, both observed: synthesis-budget overflow on very long sources (Mode A) and training-prior confabulation once the grounding source is removed (Mode B), the latter targeted by a quote-grounded refinement variant (§5).

Contributions. (1) A 21-cell same-weights measurement of end-to-end vs. two-pass decomposition on long-form review, across two modalities, three vendors, and two judges (§2, §4). (2) A nine-experiment mechanistic study that adjudicates among perception, substrate, and generation-side accounts, ruling out the first two—including behavioral substrate probes on the frontier models themselves—and supporting **perceive–externalize–synthesize** (§3). (3) A two-mode account of where decomposition breaks, with deployment triggers derived from the mechanism, plus two tested interventions and a quote-grounded iterative-refinement variant (§5). (4) Systematic robustness checks—inter-judge replication on all 21 cells, probe re-extraction, reference regeneration, $n=5$ multi-seed verification—with all retract/preserve verdicts reported transparently (§6).

2 The observation: end-to-end long-form review satisfices

Task and protocol. The task is long-form review: given a 26–85-minute recording or a 9–176-page scanned paper, write a review article that is *exhaustive* (covers the source’s content) and *faithful* (adds nothing the source does not support). For each source, a reference transcript is produced by Gemini 3.1 Pro ASR/OCR with chunking (30-min audio chunks, 1-page-per-call OCR; Appendix B). GPT-5.4 (reasoning_effort=high) acts as judge: from the reference it extracts 30–50 atomic factual probes, then scores each review for *probe coverage* (each probe COVERED/MISSING) and lists every claim the reference does not support (*hallucination count*). We treat $|\Delta h| \leq 1$ on low-count cells and $|\Delta_{\text{cov}}| \leq 0.04$ as within noise, and report Wilson/bootstrap CIs and sign tests for aggregates (Appendix B). The suite has 21 Gemini cells: 11 audio cells over 10 recordings (talks, lectures, panels, meetings) and 10 papers (seven

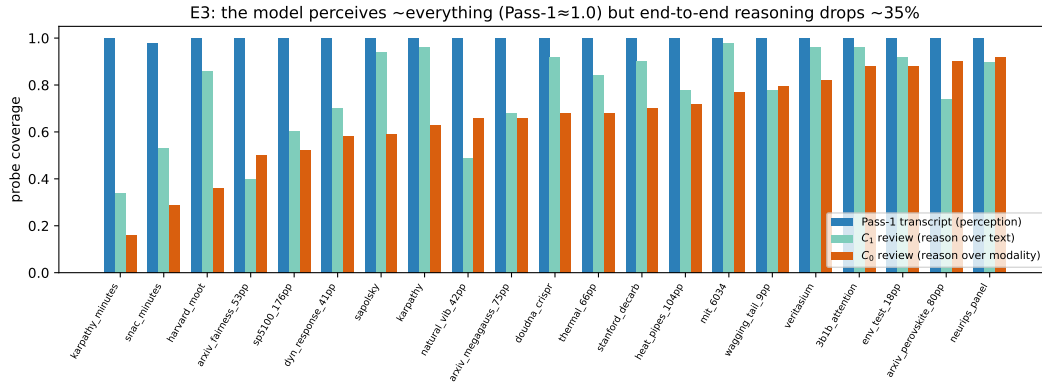


Figure 4. E3: the bottleneck is not perception. Per cell (sorted by C_0 coverage): the model’s own transcript of the source (blue) covers ≈ 1.0 of probes everywhere, while the end-to-end review (orange) surfaces 0.65 on average and the two-pass synthesis (green) 0.77. Pooled, 361/362 probes the end-to-end pass drops (99.7%, Wilson 95% CI [98.5%, 99.9%]) are present in the model’s own transcript.

1968–1972 NASA reports, three post-cutoff 2026 arXiv reviews); Appendix C lists them all. The configuration under study first is C_0 , *end-to-end*: the model reads the raw modality and writes the review in one call.

The end-to-end pass drops a third of the content... Figure 8 (gray points) shows the C_0 picture across the suite: mean probe coverage is 0.65, and every cell carries unsupported claims—most frontier reviews fail the zero-hallucination bar. The running example (Figure 3) shows what this looks like concretely: invented precision (“10,000 tokens”), structural misdescription (RLHF as a fourth stage), and silently dropped content (the talk’s closing demonstration).

...that it demonstrably perceived (E3). The natural suspicion is perception: perhaps the model cannot hear or read the source well enough. It can. We asked the same model to simply *transcribe* each source (the same chunked ASR/OCR that builds the reference) and scored that transcript against the same probes. Figure 4 shows the result: the model’s own transcript covers essentially every probe on every cell, sitting far above the review it writes from the same input. Pooled across the suite, 361 of the 362 probes the end-to-end pass drops—99.7%—are present in the model’s own transcript. On Karpathy it is 19 of 19. The transcript ceiling is partly definitional (probes come from a near-identical reference), but the load-bearing inference is one-directional: the content the end-to-end review drops *is* capturable as text by the same weights. Whatever is lost is lost *downstream* of perception. That is the puzzle the next section resolves.

3 Mechanism: why one pass fails

Three hypotheses are consistent with §2 but make different predictions. **H1 (substrate):** the model reasons better over text tokens than over raw audio/image tokens. **H2 (generation under load):** a single pass cannot simultaneously perceive, reason, and write a long faithful review, so it sacrifices; each step needs a dedicated, full-budget pass. **H3 (attention dilution):**

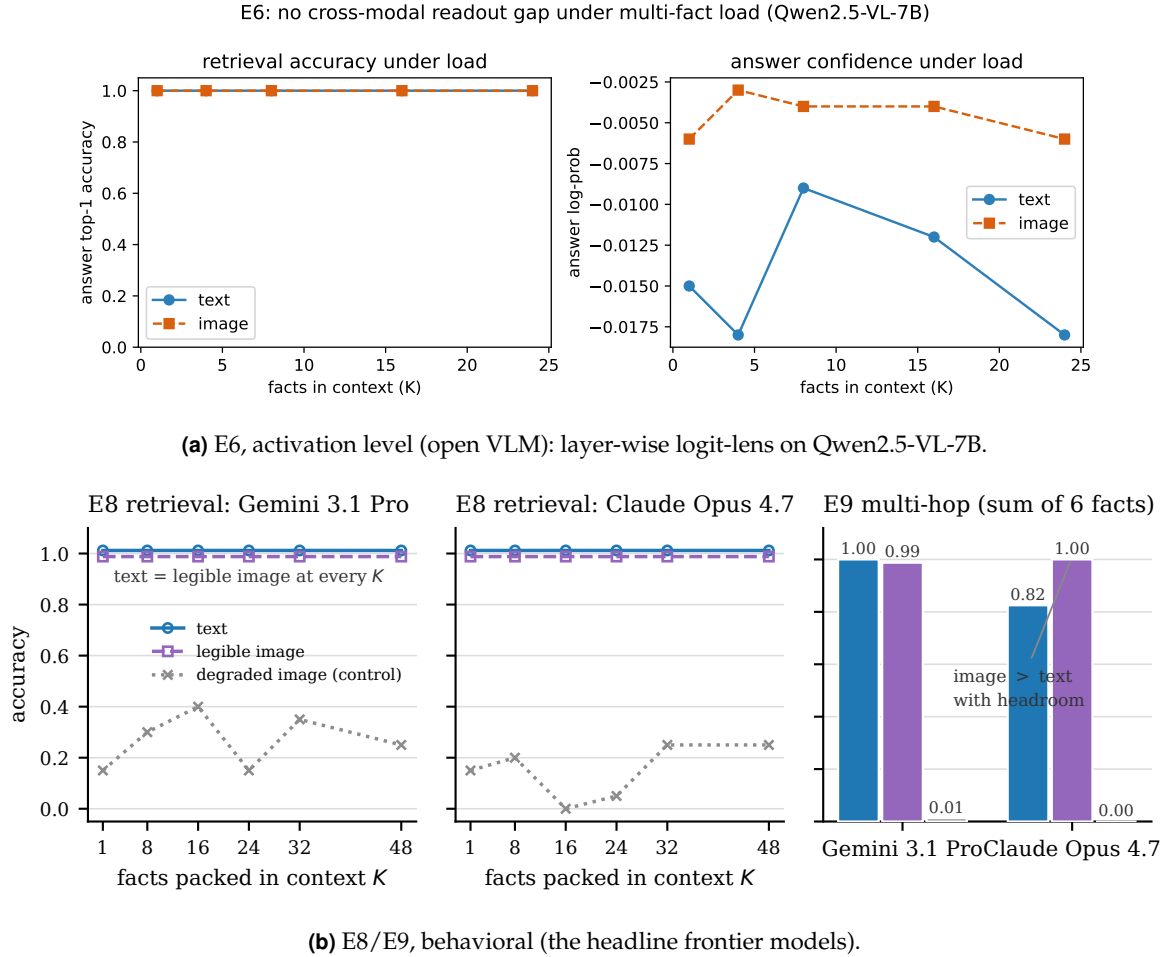


Figure 5. Text is not a better reasoning substrate than pixels (H1 is false for legible content). (a) E6: novel synthetic facts presented as text vs. as an *image* of that text are read with the same accuracy and the same answer-token log-prob at every load $K \leq 24$; readout depth is TOST-equivalent within ± 1 layer (replicated on Qwen2.5-VL-3B; Appendix J). An omni-model audio analogue (E7) gives the same null once the TTS is intelligible (Appendix A). (b) The null holds behaviorally on the exact headline frontier models. E8 (left, center): single-fact retrieval from packed sets of up to $K=48$ novel facts is *identical* from text and legible image on both models (pooled gap $+0.000$, TOST-equivalent within ± 0.1), while a degraded low-DPI control collapses to 0.15–0.27—the probe detects gaps when they exist. E9 (right): multi-hop reasoning (sum 6 packed facts; the answer is never visible). Gemini is at ceiling in both modalities; Claude without extended thinking has headroom (text 0.825) and the image is reliably *better* (1.00, gap $+0.175$, 95% CI $[0.10, 0.26]$)—the opposite of H1’s prediction.

raw-modality tokens are numerous and low-density, diluting attention. We ran nine experiments (E1–E9) to separate them, reusing the probe/hallucination judge of §2 throughout; all artifacts are released (§P). The verdict: H1 fails every test we can run, H2 is supported by the two load-bearing contrasts (both multi-seed on all 21 cells), and H3 is untestable on Gemini with our tools (its image pipeline normalizes DPI away, E4). Details and per-experiment tables are in Appendix A.

Not substrate (E6–E9). The intuitive explanation—text is intrinsically easier to reason over than pixels or audio—is directly contradicted at two levels (Figure 5). At the *activation* level,

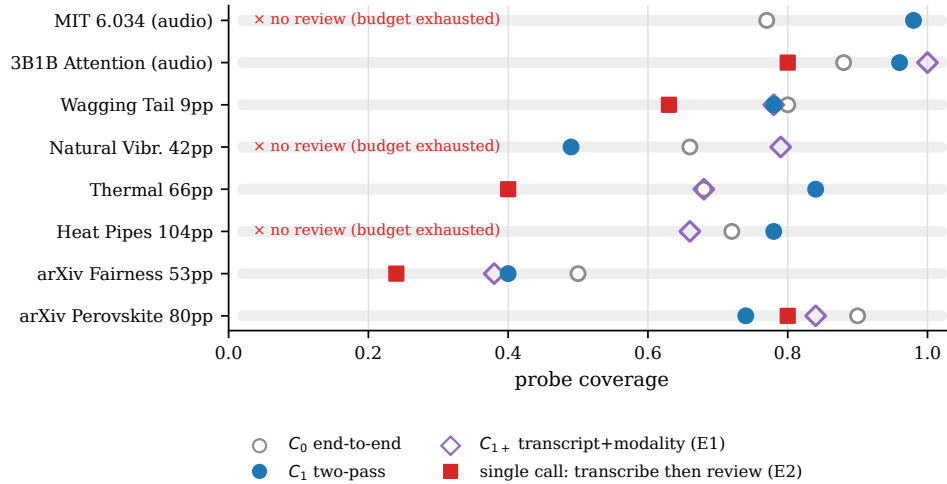


Figure 6. E2: the win needs two passes with independent output budgets; E1: removing the modality is not the engine. Probe coverage on the 8-cell condition grid (single-seed; raw counts in Appendix A, Table 5). A single call instructed to transcribe *then* review (red squares) collapses below the two-pass configuration on every paper that produced a review, and on three cells emits no review at all—the transcript exhausts the output budget. At $n=5$ seeds on all 21 cells the collapse replicates: single-call coverage sits below two-pass by $\Delta_{\text{cov}} = -0.111$ (bootstrap 95% CI $[-0.170, -0.038]$, Wilcoxon $p=0.019$, mixed-effects $p<10^{-3}$), with 7 runs emitting no review. Conversely, giving the synthesis pass the transcript *and* the raw modality (C_{1+} , open diamonds) preserves and sometimes maximizes coverage; at $n=5$ on all 21 cells it shifts coverage by only $+0.069$ (CI $[-0.001, +0.154]$, inconclusive) while re-adding hallucinations on 14/21 cells—the Mode-B effect of §5.

a layer-wise logit-lens on two open VLMs reads novel synthetic facts equally well from text and from an image of that text, for one fact and for up to 24 packed facts (E6); an omni-model analogue finds the same for speech once the TTS is intelligible—an apparent audio gap with robotic TTS vanishes with two independent clean neural voices, so audio-substrate claims must control for intelligibility (E7). *Behaviorally on the headline models*, retrieval from pixels equals retrieval from text at every load up to 48 packed facts, and a degraded-image positive control confirms the probes are sensitive (E8). Forcing multi-hop *reasoning* (E9) sharpens the contradiction: on the one model with headroom, the image is reliably better than text, not worse. The frontier probes are behavioral, not activation-level, and Gemini’s reasoning probe sits at ceiling, so Claude supplies the single headroom case; with those caveats, there is no “pixels are harder to reason over” penalty for legible content anywhere we can measure. The contrast with §2 is the point: single-fact retrieval from pixels is *perfect* on these same models, yet their end-to-end reviews drop a third of the content. The loss is not in reading the modality.

It is generation under load (E2, E1). If externalizing content into context were sufficient, one call that first transcribes verbatim and then writes the review—everything in a single context window—should match the two-pass decomposition. It does not (Figure 6): the single call collapses, either compressing the review far below the two-pass level or exhausting its entire output budget on the transcript and emitting no review at all. The complementary control (E1) shows the flip side: giving the synthesis pass *both* the transcript and the raw modality does not break the benefit—pixels in context are not harmful to coverage, consistent with

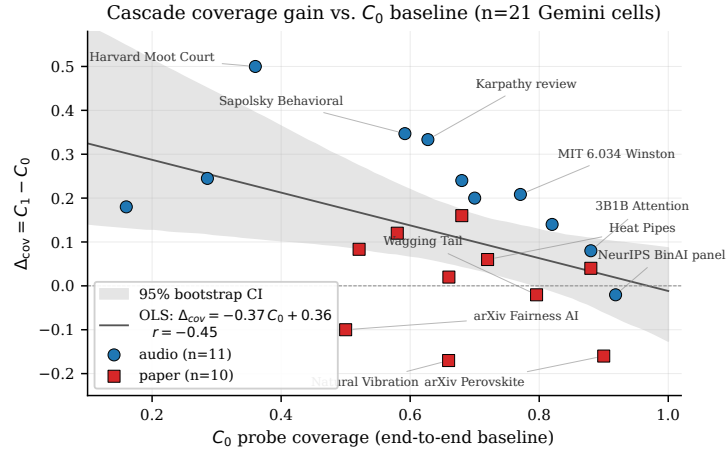


Figure 7. The inverse-baseline law. Coverage gain vs. C_0 baseline across all 21 cells: the worse the model does end-to-end, the more decomposition recovers. Circles: audio; squares: paper. OLS line with bootstrap 95% CI band; Pearson $r = -0.45$, bootstrap CI $[-0.73, -0.10]$ (fraction $r \geq 0$: 0.9%). Excluding the two meeting-minutes re-prompts (whose atomic-fact probes depress absolute coverage under a structured-minutes prompt), $r = -0.47$ on $n=19$ with CI $[-0.83, +0.10]$: the direction survives but that subset’s CI just crosses zero. The five counter-cells sit at high baseline (no headroom) or trigger Mode A/B (§5).

E6—but their presence re-introduces some hallucinations relative to pure-text synthesis. So the active ingredient is the *decomposition into two passes with independent output budgets*; removing the modality buys faithfulness only at the margin. Two further controls did not separate the hypotheses and are reported with their confounds in Appendix A: DPI cannot move Gemini’s vision-token count (E4), and pdf to text scrambles two-column reading order (E5).

Verdict: perceive, externalize, synthesize. The evidence converges on H2. The model perceives the content (E3); the modality is not the reasoning bottleneck for legible content, at the activation level and behaviorally on the frontier models (E6–E9); and recovery requires two passes with independent budgets (E2), not merely text in context (E1). One pass asked to do everything satisfies under generation load; a dedicated externalization pass followed by a dedicated, full-budget synthesis pass over complete text does not. This account makes two predictions we test next: the benefit should scale with how much the end-to-end pass satisfies (an *inverse-baseline law*, §4), and decomposition should fail in exactly two ways—when synthesis cannot fit the transcript through its output budget, and when removing the grounding source lets synthesis confabulate from training priors (§5).

4 The solution: two-pass decomposition, and where it wins

The configuration. Figure 2 formalizes the fix that §3 dictates. C_1 , the *two-pass decomposition*: Pass-1 transcribes the source to text (perceive and externalize; same prompt as the reference pipeline); Pass-2 writes the review from that transcript alone, with the same weights and the raw modality removed from context (synthesize). The only difference from C_0 is whether the reasoning-and-writing step sees pixels/audio or text. Variants used later: C_{1c} (chunked Pass-2, with `concat` for chunked-no-merge), C_1^{strip} (citation surface forms replaced by [REF]

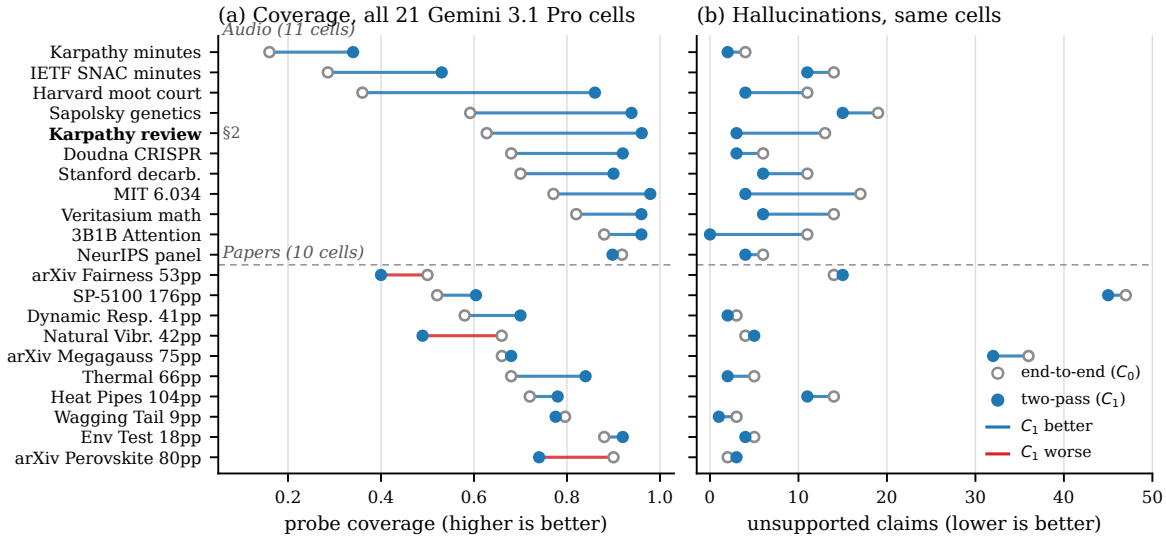


Figure 8. Per-cell results, all 21 Gemini 3.1 Pro cells (sorted by end-to-end baseline within each modality; the bolded row is the running example). Decomposition improves coverage on 16/21 (mean $\Delta_{\text{cov}} = +0.118$, bootstrap 95% CI $[+0.048, +0.189]$, sign-test $p=0.027$; 14/17 positive outside the ± 0.04 noise floor, $p=0.013$) and cuts unsupported claims on 18/21 (mean $\Delta_{\text{halluc}} = -3.86$, CI $[-5.62, -2.33]$, $p=0.0015$). Red connectors mark regressions. Raw counts and Wilson CIs in Appendix C.

before Pass-2), $C_{1\text{iter}}$ (quote-grounded iterative refinement, §5), and C_{1+} (transcript + modality, the E1 control).

Headline result at $n=21$. Figure 8 gives the per-cell picture behind Figure 1: decomposition cuts hallucinations on 18/21 cells and improves coverage on 16/21, both sign-test significant, despite Pass-2 seeing strictly less than the end-to-end pass; the direction survives restriction to cells outside the noise floor and median substitution on the four high-count judge-spread cells (§6). The largest coverage gain is Harvard Moot Court (+50pp, multi-speaker legal argumentation where the end-to-end baseline is weakest); the cleanest faithfulness win is 3B1B Attention (11 hallucinations to 0). One structural caveat qualifies the headline and is examined in §6: the reference transcript and Pass-1 share a model, which can favor C_1 ; a reference-bias control with independent ASR/OCR (Whisper, EasyOCR) preserves the direction on 4/4 cells, but that control is small, so we flag the shared-model reference as the principal threat to the headline statistic.

The inverse-baseline law. The organizing regularity is Figure 7: coverage gain scales inversely with the end-to-end baseline. This is exactly what perceive–externalize–synthesize predicts. C_0 coverage measures how much the monolithic pass satisfies on a given cell; E3 says the dropped content is sitting in the transcript; so the cells with the most headroom are the cells where a full-budget synthesis pass over that transcript recovers the most. It is a statement about *headroom*, not a promise of a universal win: the five counter-cells sit either at saturated baselines (NeurIPS panel at $C_0=0.92$; Wagging Tail ≈ 0.80) or in the two predicted failure regimes (§5), and Perovskite’s -16pp loss shrinks to -6pp —near the noise floor—under $n=5$ multi-seed re-runs (§6).

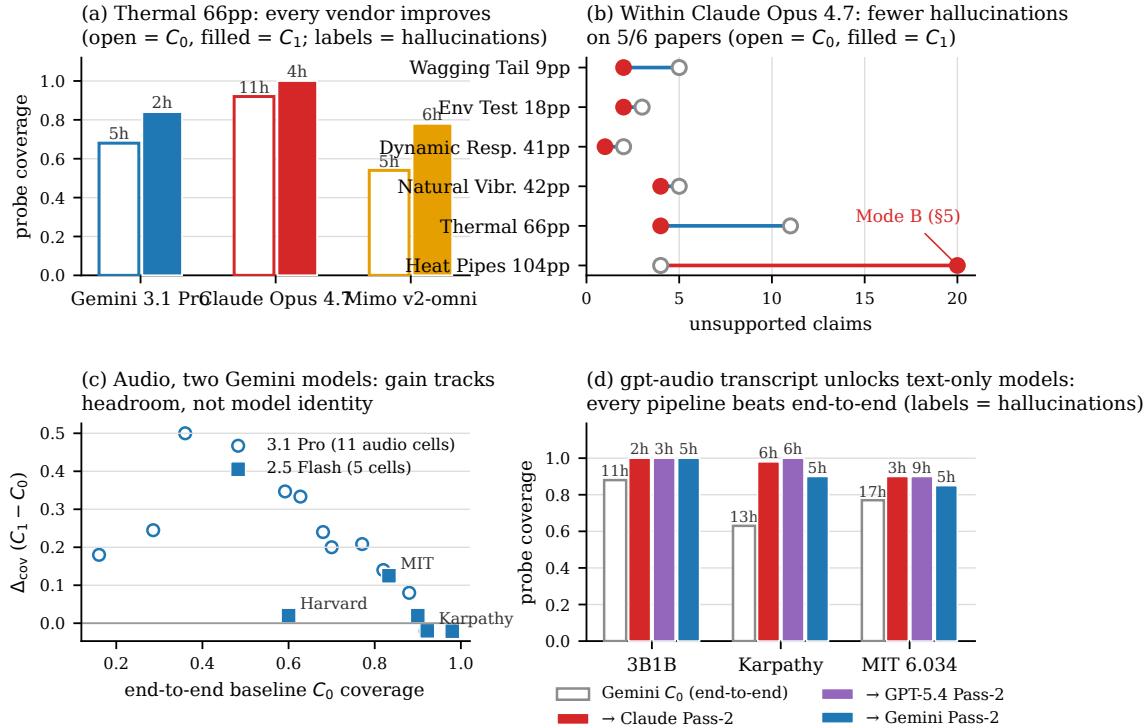


Figure 9. Cross-model generality: the inverse-baseline law, not a uniform replication. (a) Thermal 66pp, the one moderate cell all three vendors realize: every vendor improves coverage, and Gemini/Claude converge from different end-to-end hallucination rates to a common ≈ 1.4 per 1k words (Appendix N). (b) Complete within-Claude comparison on the six NASA papers its multimodal C_0 fits: hallucinations fall on 5/6 (not independently significant at $n=6$, $p=0.22$; the four short papers sit near C_0 ceiling so coverage is flat); the exception is the Heat Pipes Mode-B cell (§5). Claude C_0 is unrealizable beyond 104pp (payload cap). (c) Gemini 2.5 Flash is a *strong* end-to-end audio reader (its C_0 beats Pro’s on 4/5 cells), so its cascade gains appear only on the two headroom cells (MIT, Harvard) and Karpathy regresses—headroom-gated, as the law predicts. (d) Off-protocol mixed pipelines: an independent gpt-audio transcript feeding text-only Pass-2 models. Every pipeline beats the native audio model’s end-to-end pass on both axes; decomposition *unlocks* text-only frontier models (Claude, GPT-5.4) for audio review. Full tables: Appendices D–F.

Cross-model and cross-vendor. Every cross-model result is an instance of the same law (Figure 9): wins where the end-to-end baseline leaves headroom, washouts where it is already strong. We emphasize what this is *not*: not a blanket vendor-independent win, and the headline aggregate remains single-vendor (Gemini). A clean cross-*company* same-weights audio cascade is impossible by API constraint (no non-Google frontier model both ingests audio and accepts a text-only Pass-2; §8), which is why the mixed-model pipelines of Figure 9d are the deployment-relevant form: in production one mixes the best model per stage, and there decomposition is what makes audio review possible at all for text-only models. On long papers the vendors diverge sharply—Gemini stays small-Pareto-positive on Heat Pipes 104pp while Claude regresses on faithfulness, and on SP-5100 176pp both vendors’ C_1 show the same bibliographic-expansion signature (Appendix D)—which is the failure-mode structure the next section dissects.

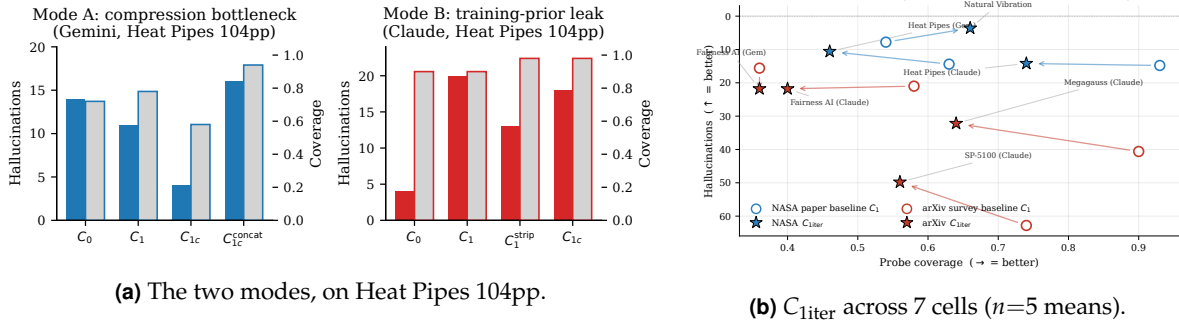


Figure 10. Failure modes and the quote-grounded fix. (a) Left: Mode A (Gemini): Pass-2 compresses a 22k-word transcript $\sim 11\times$ into a 2k-word review; chunked Pass-2 (C_{1c}^{concat}) relieves the bottleneck, lifting coverage $0.78 \rightarrow 0.94$ at +5 hallucinations. **Right:** Mode B (Claude): C_1 hallucinates several-fold more than C_0 ($4 \rightarrow 20$ on the plotted seed; $n=5$ mean 13.2), dominated by bibliographic expansions; citation stripping’s apparent fix ($20 \rightarrow 13$) is *retracted* at $n=5$ ($\hat{\Delta} = -1.8 \pm 2.6$, CI $[-6.9, +3.3]$; Appendix K). **(b) Iterative refinement:** each cell’s plain C_1 (open circle) connected to its C_{1iter} result (star). A Pareto win only on Natural Vibration ($7.8 \rightarrow 3.6h$, $0.54 \rightarrow 0.66$ cov); a faithfulness-for-coverage trade on the heavy-Mode-B cells (Heat Pipes Gemini, SP-5100, Megagauss); net-negative where Mode A dominates (Fairness AI) or the single-seed gain washes out (Heat Pipes Claude). Per-cell table: Appendix G.

Cost. C_1 is $N+1$ calls versus one: call ratios range from $\approx 2\times$ on short audio to $\approx 177\times$ on the 176pp paper, with output-token volume scaling with source length (cost-vs-gain scatter in Figure 12, §8). The evidence says decomposition pays off where end-to-end has measurable headroom and buys nothing on saturated cells—so the inverse-baseline law is also the cost rule.

5 Where decomposition breaks: two failure modes and their fixes

The mechanism predicts exactly two ways decomposition can fail, and both appear. **Mode A (compression bottleneck):** Pass-2 has a fixed output budget ($\approx 2k$ words); when the transcript is many times longer, synthesis must compress and drops content the same way C_0 does. **Mode B (training-prior leak):** removing the raw source removes a grounding anchor; on documents with heavy training-data overlap, Pass-2 fills abbreviated details from prior knowledge. Mode B is the genuine—but secondary—cost of dropping the modality (it is the same effect E1 shows in reverse). Figure 10a shows both on Heat Pipes 104pp, the cell where they separate cleanly by vendor.

Mode A: compression bottleneck (3 papers). Three Gemini cells show the signature of transcript-length-to-budget ratios of $\sim 9\text{--}15\times$: Heat Pipes 104pp (small +6pp gain), arXiv Fairness AI 53pp (−10pp), arXiv Perovskite 80pp (−16pp at $n=1$, −6pp under multi-seed). The tested fix is **chunked Pass-2**: split the transcript into $\sim 5k$ -word chunks, summarize each, concatenate. It lifts Heat Pipes coverage $0.78 \rightarrow 0.94$ (Figure 10a) and replicates directionally on Fairness AI and Perovskite at $n=5$ but *not* on Megagauss, so we report chunked Pass-2 as a 2/3-replicated Mode-A fix (Appendix H).

Mode B: training-prior leak (4 papers). The Heat Pipes paper cites “Reference 2 (Cotter)”; Claude’s C_1 expands this to “Cotter’s 1965 analysis (LA-3246-MS)” —the year and report num-

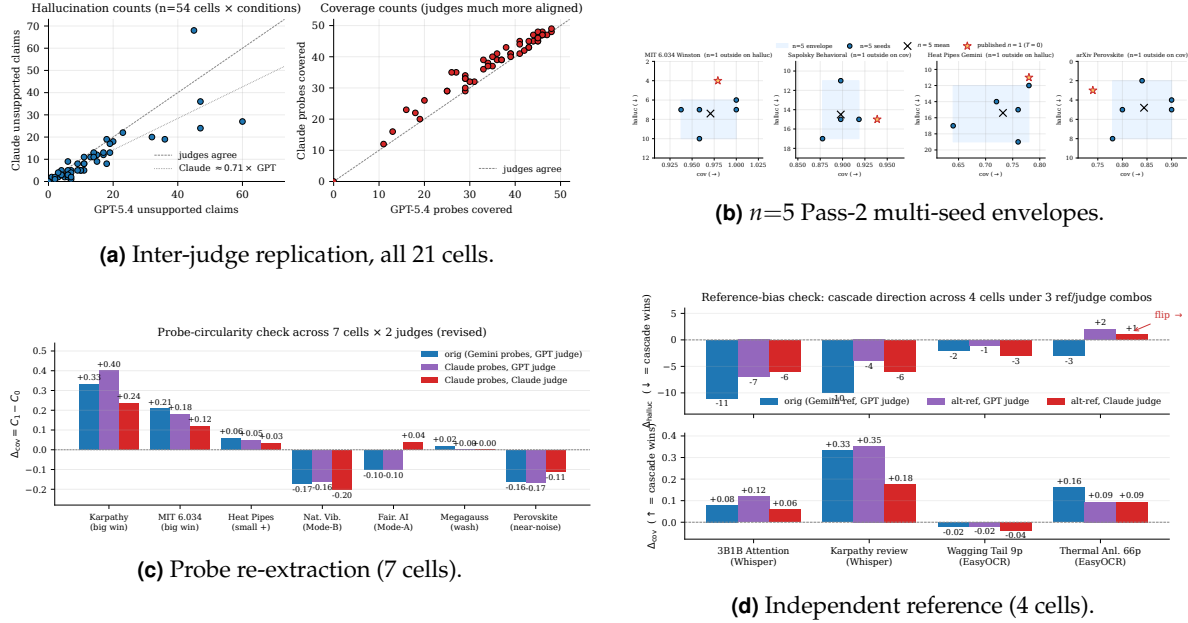


Figure 11. Four independent robustness checks (full tables: Appendices I–M). **(a)** Claude Opus 4.7 re-scores every review with GPT-5.4’s prompts: direction agreement 19/21 on coverage, 14/21 on hallucination, per-probe agreement median 0.94; Claude reports systematically fewer hallucinations (median ratio $0.71\times$), so Mode B holds under both judges at different absolute counts. Disagreements concentrate on near-tie and high-count cells. **(b)** Five-seed Pass-2 re-runs on four headline cells: the three wins preserve direction; Perovskite’s loss shrinks from -16pp to -6pp , below the noise floor. **(c)** Probes re-extracted by Claude on seven cells spanning the direction spectrum: cascade Δ_{cov} sign preserved on 6/7 under the GPT-5.4 judge and 5/7 under Claude (the exceptions are within-noise cells). **(d)** Reference regenerated with independent ASR/OCR (Whisper-base, EasyOCR): coverage direction preserved 4/4 under both judges; the one hallucination-direction flip (Thermal under EasyOCR) traces to the noisier reference losing figure captions.

ber are training-data details absent from the transcript (more examples in Appendix O). The same bibliographic-expansion signature appears on all four long, citation-heavy Claude cells (Heat Pipes, SP-5100, Fairness AI, Megagauss) and is absent on the citation-light Thermal 66pp. Two deployment triggers emerge (thresholds and per-cell citation densities in Appendix C): Mode A when the transcript is $\gtrsim 5\times$ the Pass-2 output budget and the coverage gain is small; Mode B when the source carries $\gtrsim 2$ citation surface forms per 1k words *and* its topic is in widely cited pre-cutoff literature. The modes compound (Mimo Heat Pipes shows both) or occur alone (Heat Pipes: Gemini dominantly A, Claude dominantly B).

Interventions, including a retraction. Beyond chunked Pass-2, we tested *citation stripping* (C_1^{strip}) as a Mode-B fix. **It does not survive replication:** the original “20 $\rightarrow$ 13” Heat Pipes point estimate lies within the combined Pass-2+judge noise envelope at $n=5$, and across four cells the fix has zero replicated wins (Appendix H). We retract it, and it motivates the stronger structural intervention below.

C_{liter} : a quote-grounded refinement. Mode B calls for re-grounding synthesis in the transcript. C_{liter} : (a) extract 50–150 atomic claims from the transcript, each with a verbatim ≤ 25 -

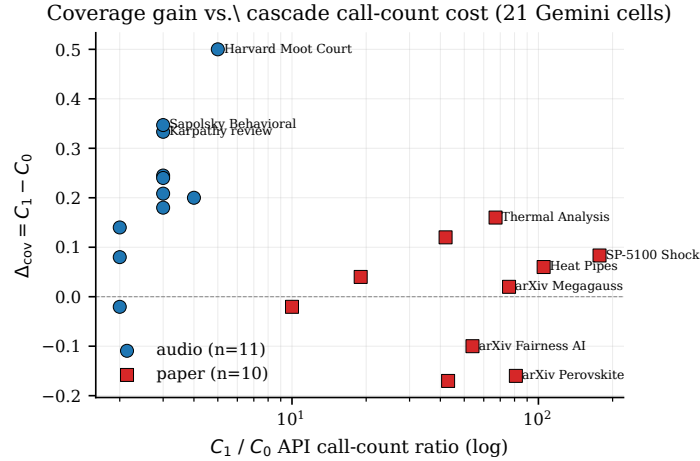


Figure 12. Coverage gain vs. API call-count cost. Each point is one Gemini cell. The C_1/C_0 call ratio for paper cells is $N+1$ (page-chunked OCR + one Pass-2), from $10\times$ on a 9-page paper to $177\times$ on SP-5100; audio cells use 30-min chunked ASR ($\approx 2\text{--}3\times$). Decomposition buys most of its coverage gain at the cheap audio end and small or negative gains at the high-cost paper end—e.g. Perovskite at $81\times$ loses coverage (-6pp under multi-seed), while SP-5100 at $177\times$ still gains $+8\text{pp}$ because its C_0 baseline (0.52) leaves so much headroom. The inverse-baseline law is also the cost rule.

word quote; (b) drop any claim whose quote does not fuzzy-match the transcript; (c) rewrite the review from surviving claims only. The filter is a tripwire, not a culler—99.8% of claims pass (3731/3738)—so its effect is to make the model self-discipline: asked for a quote, it stops emitting the bibliographic expansions Mode B catalogues. It is RAG’s attribution discipline applied *within* a same-weights self-cascade. The verdict at $n=5$ on seven cells (Figure 10b) is conditional on the dominant failure mode: it suppresses Mode B where Mode B binds, pays a rewrite-compression cost everywhere, and is therefore a Pareto win only where the rewrite does not over-compress.

6 Robustness

Figure 11 summarizes four checks against the protocol’s main threats: judge idiosyncrasy (a), Pass-2 and judge stochasticity (b), probe circularity—GPT-5.4 extracts and also scores the probes (c)—and the shared-model reference (d), which is the structural caveat of §4 and the one we consider the principal residual threat: the 4/4 preservation is reassuring but the control is small and concentrated on strong-win cells.

Separating stochasticity sources (Figure 13, §8; Appendix K) yields three operational facts. Pass-2 generation varies (Claude deprecates `temperature`, so determinism is not enforceable; Gemini at $T=0$ is mostly but not strictly deterministic); the GPT-5.4 judge is essentially deterministic on low-count cells but spreads $\pm 10\text{--}25$ on cells with very long unsupported-claim lists (which run near its completion budget); and coverage is far more stable than hallucination counts under all three sources. Accordingly, headline aggregates use median substitution on the four high-count cells, hallucination counts above ~ 30 carry judge-side uncertainty, and—the pattern worth internalizing—the Mode B *observation* survives every noise source while the Mode B *fix* (citation stripping) did not, which is what forced the retraction in §5. Finally, a leave-one-out logistic predictor of $\text{sign}(\Delta_{cov})$ from four cell features reaches LOOCV

Table 1. Practitioner checklist drawn from this paper’s cells: which configuration to deploy per source/model regime.

Source / model class	Diagnosis	Recommended action
Audio talk or moderate paper (40–70pp) on a capable model	Cascade strongly Pareto-positive.	Default C_1 .
Moderate paper with Mode-B Gemini signature (Natural Vibration 42pp)	Mode B dominant, and the cell flips to Pareto-positive under $C_{1\text{iter}}$.	$C_{1\text{iter}}$.
Long paper, Mode A dominant (Heat Pipes 104pp Gemini)	Compression bottleneck.	C_{1c}^{concat} (chunked, no merge).
Long paper, Mode B dominant (Heat Pipes Claude)	Citation-prior leak, but the $C_{1\text{iter}}$ faithfulness gain washes out at $n=5$.	C_0 if it fits. $C_{1\text{iter}}$ buys only a coverage loss.
Long arXiv survey (Fairness AI, SP-5100, Megagauss)	Mode A + Mode B compound. At $n=5$, $C_{1\text{iter}}$ is net-negative on Fairness AI but a faithfulness/coverage trade on SP-5100/Megagauss. Chunked Pass-2 partially replicates.	Prefer C_0 when it fits. If cascade required, use C_{1c}^{concat} with multi-seed averaging.

AUC 0.65 with all coefficient signs matching the qualitative picture—an honest $n=21$ baseline, well short of deployment-ready (Appendix C).

7 Related work

Speech and documents. Prior cascade-vs-end-to-end results are often read as substrate claims; our mechanism study suggests they are at least partly about decomposition and modality-specific recoverability. Cuervo et al. [2] document a text/speech understanding gap, and the Cascade Equivalence Hypothesis [3] shows speech-LLMs *internally* behave like ASR→LLM cascades, while X-Talk [4] finds explicitly cascaded speech-to-speech pipelines competitive. On documents, DocVLM [6] reports OCR-augmented VLMs beating pure-vision VLMs on DocVQA at fixed budget, while the OCR-Reasoning Benchmark [7] shows pure-OCR cascades cap below 50% where spatial layout carries meaning, ColPali [8] shows vision-only retrieval beating OCR, and GUI agents do better from raw pixels than from structured-text intermediates [9, 10]. The common thread matches our mechanism: decomposition helps when the task is generation-under-load and the content survives transcription, and hurts when the answer lives in layout or signal that transcription discards.

Step-Audio-R1: a boundary, not a contradiction. Step-Audio-R1 [5] observes audio LLMs spontaneously reasoning over a *textual surrogate* of the audio (attributing musical sadness to “lyrics mention sadness” rather than a minor key) and treats this as the failure its modality-grounded distillation must correct. Their tasks—music, environmental sound, paralinguistics—have *acoustic* answers that transcription discards; ours is the complementary regime, where the payload is what was *said*. Our activation tests find no reasoning-substrate gap for propositional content in either modality (E6/E7), so we read their effect as task-payload-specific rather than as “audio is a worse substrate.” Our Mode B is the within-regime echo of their concern.

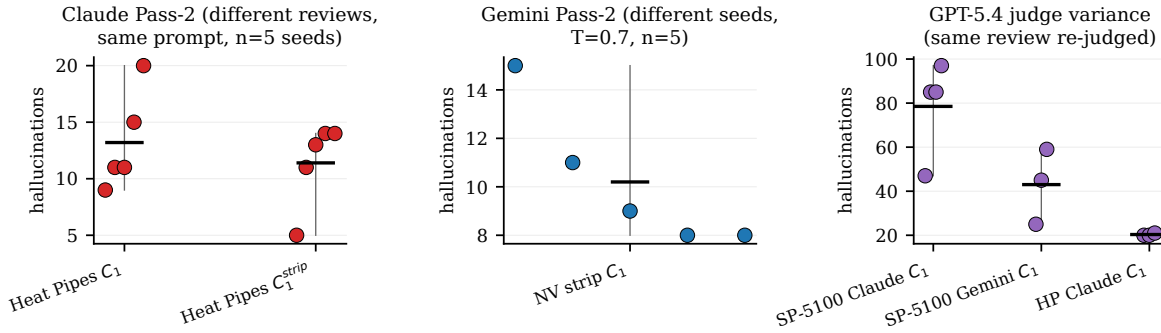


Figure 13. Three sources of stochasticity, decomposed. *Left:* Claude Pass-2 generation variance, with 5 reruns on Heat Pipes yielding {9, 11, 11, 15, 20} (range 11). *Middle:* Gemini Pass-2 generation variance at $T=0.7$ on Natural Vibration strip: {15, 11, 9, 8, 8} (range 7). *Right:* GPT-5.4 judge variance on the *same* review file judged multiple times. SP-5100 Claude C_1 produces {47, 85, 85, 97} over 4 judge re-runs (range 50), while Heat Pipes Claude C_1 is stable across re-judgments. Coverage is much more stable than hallucination counts across all three noise sources.

Cascade theory and faithfulness. Bengio’s Consciousness Prior [11] argues for sparse discrete bottlenecks between perception and reasoning; our externalized text bottleneck is a deployment-time analogue, and Discrete JEPa [12] makes a related architectural argument within vision. Mode B is a self-cascade analogue of the unfaithfulness problem motivating RAG [13], RARR [14], and Attribution to Identified Sources [15]; $C_{1\text{iter}}$ adopts RARR’s in-context-attribution discipline but operates entirely within a same-weights self-cascade without external retrieval.

8 Discussion and limitations

What generalizes. The portable result is not “decomposition always wins” but the inverse-baseline law: decomposition recovers content in proportion to how much the end-to-end pass sacrifices, so it pays off precisely where a model reasons poorly over the raw modality and buys nothing where it is already strong. This is what makes the cross-vendor picture coherent rather than contradictory—Claude’s near-ceiling NASA papers, Gemini 2.5 Flash’s strong end-to-end audio, and Gemini Pro’s weak end-to-end audio all fall on the same line (Figure 9). The most deployable corollary is that decomposition *unlocks* text-only frontier models for audio (Figure 9d). Table 1 compresses the per-regime recommendations (C_1 , $C_{1\text{iter}}$, C_{1c}^{concat} , or C_0).

Limitations. *Scope:* the headline aggregate is single-vendor; the within-Claude comparison replicates the faithfulness direction on 5/6 papers but is not independently significant at $n=6$, and Claude C_0 is unrealizable beyond 104pp. A cross-company same-weights *audio* cascade is API-impossible (Claude takes no audio; OpenAI’s audio model refuses text-only Pass-2; both verified), so the mixed pipeline is the natural production form. The Mimo audio cells ran at reduced bitrate (OpenRouter cap), conflating quality with model weakness; the full-quality 2.5 Flash arm removes that confound and is headroom-gated as predicted. *Measurement:* probe and reference bias are only partially controlled (7 and 4 cells); the shared-model reference remains the principal threat to the headline statistic; single-judge intra-rater

variance affects high-count cells ($\pm 10\text{--}25$; Figure 13). Several sources predate the model cutoff, but memorization biases *against* the cascade (it aids C_0), making the headline conservative; the post-cutoff sources span wins and counter-cells, consistent with the law. *Mechanism breadth*: all measurements are on long-form review—we claim nothing about short-form QA, code, or agentic tasks. E8/E9 are behavioral, not activation-level (frontier activations are inaccessible); Gemini’s reasoning probe sits at ceiling, so Claude supplies the one headroom case; real scanned/degraded pages remain untested (the degraded control is synthetic); E4/E5 are single-seed and confounded.

Validation plan. The load-bearing effects are already multi-seed on all 21 cells, so remaining replication is about breadth, not statistical power: extend the E6 logit-lens to other open families and, where activations become accessible, lift E8/E9 to the activation level; push the audio analogue (E7) into the multi-fact load regime; re-test on real scanned and degraded pages, where a substrate or dilution effect is most likely to appear; extend to other generation-under-load tasks plus a short-form control where the account predicts *no* decomposition benefit; and causally vary the synthesis output budget to confirm Mode A is budget-bound.

9 Conclusion

On long-form review, the intuitive end-to-end deployment of a multimodal model quietly satisfies: it drops a third of what the model demonstrably perceived and embellishes the rest. The cause is not perception and not the reasoning substrate—text and images of text are read equally well, on open models at the activation level and on the frontier models behaviorally—but generation under load: one pass cannot perceive, reason, and write a long faithful review at once. A same-weights two-pass decomposition with independent output budgets recovers the loss, beats end-to-end on faithfulness and coverage at $n=21$, and obeys an inverse-baseline law: it helps most exactly where the monolithic pass satisfies most. Its two predicted failure modes—synthesis-budget overflow and source-removal confabulation—both appear, with a chunked Pass-2 and a quote-grounded refinement as partial, honestly-replicated fixes. The practical rule is **perceive, externalize, then synthesize**: the win is the decomposition, not the modality—a deployment-time pattern with predictable failure modes, not a universal architecture.

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A Mechanistic experiments E3–E9 and controls E4/E5, in detail

This appendix expands the mechanism sections (§2 E3, §3 E1/E2/E6–E9) with the per-experiment tables and the two confounded controls.

E3: perception is not the bottleneck. The per-cell breakdown is Figure 4 (main text). The transcript covers a mean 99.9% of probes, versus 65% for the C_0 review and 77% for C_1 synthesis; 361/362 probes dropped by C_0 are present in the model’s own transcript.

E6: no cross-modal substrate gap at the activation level. We tested H1 directly on an open VLM (Qwen2.5-VL-7B), controlling for prior knowledge with novel synthetic facts about nonsense entities (e.g. “the Zorblatt device weighs 7 kilograms”). Each fact is presented either as digital text or as an *image* of that same text, with a matched single-token question, and a layer-wise logit-lens reads off the log-probability and rank of the answer token at every layer. The two modalities are indistinguishable for an isolated fact and under load: packing $K \in \{1, 4, 8, 16, 24\}$ facts into one stimulus, image and text stay at 1.00 accuracy with a readout-depth gap < 1 layer (TOST-equivalent within ± 1 layer) at every K (Figure 5a). The same null replicates on Qwen2.5-VL-3B; the per- K ladder for both models is in Appendix J. (The 32B variant verbalizes answers in a form our single-token matcher cannot score, so it is excluded.)

E7: no audio substrate gap either, once the speech is intelligible. We replicate the E6 logit-lens on an omni model (Qwen2.5-Omni Thinker), presenting each novel fact as text vs as *synthesized speech*, across three TTS engines of increasing quality (Table 2). The result is a methodological caution turned finding: with low-quality TTS (espeak, only 71% of clips intelligible to the model’s own ASR) audio *appears* to have a substrate gap, but this tracks *intelligibility*, not modality—with two independent clean neural voices (88% and 96% intelligible) the gap vanishes and is TOST-equivalent within ± 1 layer. Garbled TTS manufactures a spurious gap, so audio-substrate claims must control for TTS intelligibility.

Table 2. E7 audio readout (Qwen2.5-Omni Thinker, 24 novel facts): the apparent audio substrate gap is a TTS-intelligibility artifact. “Intellig.”: fraction of clips the model’s own ASR transcribes correctly. Depth-gap = audio–text readout depth (layers). TOST SESOI ± 1 layer.

TTS	intellig.	text acc	audio acc	depth-gap 90% CI	TOST-equivalent
espeak (robotic)	0.71	0.96	0.83	[0.64, 1.78]	no (spurious gap)
mms-tts-eng (clean)	0.88	0.96	1.00	[0.18, 0.57]	yes
fish-speech-1.5 (natural)	0.96	0.96	1.00	[0.21, 0.54]	yes

E8: frontier behavioral retrieval probe. E6/E7 establish the substrate null only on small open models (the lens needs activations). E8 closes the gap behaviorally on the exact headline models: pack $K \in \{1, 8, 16, 24, 32, 48\}$ novel nonsense-entity facts (distinct two-digit values, so prior knowledge gives no shortcut) as text or as an image of that text, and ask a single-number question about one target entity ($T=20$ seeded trials/cell). Table 3 gives the counts behind Figure 5b: on both models, legible-image accuracy equals text accuracy exactly at every K up to 48, while the degraded positive control (small font, downscaled and blurred to low-DPI-scan quality) collapses and is decisively non-equivalent—the probe is sensitive.

Table 3. E8 behavioral cross-modal substrate probe ($T=20$ seeded trials per K , single-number retrieval). “legible img”: image of the same text at 34px. “degraded img”: small-font, downscaled+blurred low-DPI-scan control. Gap = image–text accuracy; TOST SESOI ± 0.1 .

Model	cond	acc @ $K=1$	acc @ $K=48$	pooled gap (95% CI)	TOST-equiv
Gemini 3.1 Pro	text	1.00	1.00	—	—
	legible img	1.00	1.00	+0.000 [0.00, 0.00]	yes
	degraded img	0.15	0.25	−0.733 [−0.81, −0.65]	no (control)
Claude Opus 4.7	text	1.00	1.00	—	—
	legible img	1.00	1.00	+0.000 [0.00, 0.00]	yes
	degraded img	0.15	0.25	−0.850 [−0.91, −0.78]	no (control)

E9: frontier multi-hop reasoning probe. E9 forces composition: the model must pull $M=6$ specified facts out of a K -fact packed set and *sum* them—the total is never a visible number—as text vs. legible image vs. the E8 degraded control ($T=20$ /cell over $K \in \{8, 16, 24, 32\}$, Table 4). On Gemini, text and legible image are both at ceiling and TOST-equivalent (gap −0.013). On Claude without extended thinking—the one sub-ceiling regime we could produce (text accuracy 0.825; we could not drive clean-text Gemini below ceiling at any difficulty tried)—the legible image is reliably *better* (image 1.00, gap +0.175, 95% CI [0.10, 0.26]); the 14/80 text errors are near-miss arithmetic slips that the image condition gets right. We read the Claude effect as a within-prompt echo of this paper’s thesis—reading from an image nudges the model to externalize the numbers before summing—but report only the direction.

Table 4. E9 multi-hop reasoning probe: retrieve $M=6$ packed facts and sum them (answer never directly visible), $T=20$ /cell over $K \in \{8, 16, 24, 32\}$. Gap = image–text accuracy; TOST SESOI ± 0.1 .

Model	text acc	legible-img acc (gap)	TOST-equiv	degraded-img acc
Gemini 3.1 Pro	1.00	0.99 (−0.013)	yes	0.01
Claude Opus 4.7	0.825	1.00 (+0.175)	no (image <i>better</i>)	0.00

E4, E5: controls that did not separate the hypotheses. E4 sought to vary vision-token count via DPI on a fixed document to probe H3. But Gemini normalizes images internally, so 96 and 200 DPI yield near-identical prompt token counts (112,007 vs 111,855 on Heat Pipes); the manipulation does not move the independent variable, and H3 is untestable this way on Gemini. E5 fed native digital text (pdf to text) to a single text-only call as a clean H1 test,

but on two-column arXiv PDFs `pdf to text` scrambles reading order, so native-text coverage is confounded downward (Fairness AI 0.34, below even C_0). Neither supports H1, so we report both with their confounds rather than omit them.

Table 5. Mechanistic conditions on the 8-cell subset (hallucinations h / probe-coverage); the data behind Figure 6. C_0 : end-to-end (modality). C_1 : two-pass decomposition (text). C_{1+} : synthesis pass given both transcript and modality (E1). single-call: transcribe-then-review in one response (E2, “–” = no review emitted, budget exhausted). native-text: single text-only call from `pdf to text` (E5, arXiv only). Single-seed.

Cell	C_0 (img/aud)	C_1 (text)	C_{1+} both	single-call	native-text
MIT 6.034 (audio)	17h/0.77	4h/0.98	–	– (budget)	–
3B1B Attention (audio)	11h/0.88	0h/0.96	5h/1.00	1h/0.80	–
Wagging Tail 9pp	3h/0.80	1h/0.78	2h/0.78	5h/0.63	–
Natural Vibration 42pp	4h/0.66	5h/0.49	9h/0.79	– (budget)	–
Thermal 66pp	5h/0.68	2h/0.84	1h/0.68	3h/0.40	–
Heat Pipes 104pp	14h/0.72	11h/0.78	10h/0.66	– (budget)	–
arXiv Fairness 53pp	14h/0.50	15h/0.40	11h/0.38	11h/0.24	13h/0.34
arXiv Perovskite 80pp	2h/0.90	3h/0.74	5h/0.80	5h/0.80	4h/0.80

B Protocol details

Reference quality. Gemini 3.1 Pro is used for OCR/ASR with chunking. Single-call OCR on a 104-page paper produced a degenerate TOC dot-leader loop (113KB output, 56k “words” but only 264 real-word tokens, 6 page markers out of 104). Chunking (1 page per call, max 8192 output tokens) eliminates this failure mode. For audio we chunk into 30-min segments with 5-second overlap. We discovered a silent server-side truncation on one audio chunk (chunk 1 of IETF SNAC produced 1981 words instead of the expected ≈ 4368 , ending mid-sentence with “Uh, well, see, yeah, that’ ”). The protocol grew an automatic chunk-level word-count parity check after this incident.

Statistics. We treat $|\Delta h| \leq 1$ on cells with $h < 10$, and $|\Delta \text{cov}| \leq 0.04$ (≈ 2 probes on a 50-probe cell), as within noise (presentation conventions calibrated against inter-judge direction flips, §6). For aggregate effects we report Wilson 95% CIs on coverage rates, percentile-bootstrap 95% CIs ($n_{\text{boot}}=10000$, seeded) on cross-cell means and Pearson r , and sign tests against $\Delta=0$. The judge runs with a 24k-token completion budget.

Sources. Audio: Karpathy “State of GPT” (42 min, 2023), MIT 6.034 L1 Patrick Winston (47 min, 2010), Sapolsky Behavioral Genetics 1 Stanford (85 min, post-2026 reupload), Stanford Decarbonized Power Sector Zach Ming (72 min, 2025), Doudna CRISPR Basics (48 min, 2017), 3Blue1Brown “Attention in Transformers” (26 min, 2024), Veritasium “Strange Math” (32 min, 2025), NeurIPS Black-in-AI panel (31 min, 2021), Harvard Law School Ames Moot Court (82 min, 2025), IETF SNAC WG interim 2026-04-16 (60 min, ≈ 5 speakers, recorded post the most likely training-data cutoff). Karpathy yields both a review and a meeting-minutes task. Papers: NASA NTRS IDs 19720009221 (Wagging Tail, 9pp), 19680000724 (Env Test, 18pp), 19690008405 (Dynamic Response, 41pp), 19690013408 (Natural Vibration, 42pp),

19700023812 (Thermal Analysis, 66pp), 19700025120 (Heat Pipes, 104pp), and SP-5100 (Shock and Vibration Technology, 176pp). The arXiv papers are arXiv:2605.09852 (Fairness AI, 53pp), arXiv:2605.13991 (Perovskite Tandem PV, 80pp), and arXiv:2605.11379 (Megagauss Physics, 75pp). NASA papers are rendered as page images at 200 DPI, and arXiv papers at 200 DPI full color.

Models and snapshots. The models are Gemini 3.1 Pro Preview (gemini-3.1-pro-preview), Claude Opus 4.7 (claude-opus-4-7), and MIMO v2-omni (xiaomi/mimo-v2-omni via OpenRouter). The judge is GPT-5.4 (reasoning_effort=high, completion budget 24k tokens), routed via OpenRouter due to scope restrictions on the local OpenAI key. Cross-vendor extensions reuse the Gemini reference and probes, isolating the question to whether the advantage depends on the C_0/C_1 model: Claude Opus 4.7 (complete C_0 -vs- C_1 on the six papers where its multimodal C_0 fits; C_1 -only on the longer papers) and MIMO v2-omni on every cell where its C_0 does not return empty.

MIMO audio constraint. OpenRouter’s MIMO provider enforces a 10MB base64 audio input limit. Karpathy required re-encoding from 32kbps to 16kbps mono mp3 to fit, and Sapolsky at 85 min required 12kbps. The MIMO audio Karpathy result (C_0 4h/0.569, C_1 4h/0.588 across 51 probes, 1509 / 1268 review words) is within noise across the $C_0 \rightarrow C_1$ transition, so we treat it as a wash. *gpt-audio* (via OpenRouter) accepted audio input but refused text-only Pass-2 with HTTP 400 “This model requires that either input content or output modality contain audio”. The same-weights cascade is not realizable on this model, and C_0 alone scored 10h/0.765 on Karpathy.

Pareto terminology. We call a cascade condition *strictly Pareto-positive* relative to C_0 when it has both fewer hallucinations and higher coverage by amounts exceeding the noise floor. *Pareto-positive within noise* means strictly better on one axis and no worse than noise on the other. *Pareto-dominant* on a set means no other condition has lower hallucination at equal or higher coverage. These are presentation labels, not statistical claims.

C All 21 cells: raw counts and Wilson 95% CIs

A held-out predictor (LOOCV, $n=21$). A leave-one-out logistic-regression predictor on four cell-level features (C_0 probe coverage, $\log_{10}$ source words, C_0 unsupported-claim count, and a paper/audio dummy) reaches LOOCV AUC = 0.65 on $\text{sign}(\Delta_{\text{cov}} > 0)$. All four coefficient signs match the qualitative picture (higher baseline reduces cascade benefit, longer source reduces benefit, more C_0 hallucinations *increases* benefit, paper modality harder than audio). AUC 0.65 is meaningful but well below deployment-ready, so we report it as an honest $n=21$ baseline. Adding a citation-density feature drops AUC to 0.61: the residual Mode-B structure is not captured by a single density feature at this sample size.

D Cross-vendor details

SP-5100 176pp replication. NASA SP-5100 “Shock and Vibration Technology” (1972, 176pp, 47k pdftotext words, 4.00 citation surface forms / 1k words) was chosen as a second long

Table 6. All 21 Gemini base cells (the data behind Figures 1 and 8). Aggregates: mean $\Delta_{\text{cov}} = +0.118$ (bootstrap 95% CI $[+0.048, +0.189]$), mean $\Delta_{\text{halluc}} = -3.86$ (CI $[-5.62, -2.33]$); restricting to the 17 cells outside the coverage noise floor leaves 14/17 positive (sign-test $p=0.013$). Halluc. is the raw unsupported-claim count (single judge call unless noted, with medians across multi-judge data given in brackets for high-count cells). Cov. is k/n (covered probes out of n) with Wilson 95% CI. Cells where C_1 does not strictly improve on at least one axis above the noise floor are marked with †. *cite/1k* = count of citation surface forms (numeric brackets + “Ref. N ” word forms) per 1k source (pdf to text) words—the Mode-B trigger feature of §5 (Heat Pipes: 2.80; Thermal: 0.36).

Cell	len	C_0 halluc	C_0 cov [95% CI]	C_1 halluc	C_1 cov [95% CI]	Δ_{cov}	<i>cite/1k</i>
<i>Audio (11 cells)</i>							
Karpathy review	42m	13	0.627 [0.490, 0.747]	3	0.961 [0.868, 0.989]	+0.333	0.00
Karpathy mtg-min	42m	4	0.160 [0.083, 0.285]	2	0.340 [0.224, 0.478]	+0.180	0.00
SNAC mtg-min	60m	14	0.286 [0.178, 0.424]	11	0.531 [0.394, 0.663]	+0.245	0.00
3B1B Attention	26m	11	0.880 [0.762, 0.944]	0	0.960 [0.864, 0.989]	+0.080	0.00
Veritasium Math	32m	14	0.820 [0.692, 0.903]	6	0.960 [0.864, 0.989]	+0.140	0.00
MIT 6.034 Winston	47m	17	0.771 [0.633, 0.866]	4	0.979 [0.890, 0.996]	+0.208	0.00
NeurIPS BinAI panel	31m	6	0.918 [0.803, 0.969]	4	0.898 [0.778, 0.957]	−0.020†	0.00
Doudna CRISPR	48m	6	0.680 [0.542, 0.792]	3	0.920 [0.812, 0.968]	+0.240	0.00
Stanford Decarb	72m	11	0.700 [0.562, 0.809]	6	0.900 [0.786, 0.957]	+0.200	0.00
Harvard Moot Court	82m	11	0.360 [0.241, 0.499]	4	0.860 [0.736, 0.932]	+0.500	0.00
Sapolsky Behavioral	85m	19	0.592 [0.450, 0.720]	15	0.939 [0.834, 0.980]	+0.347	0.00
<i>Paper, NASA NTRS 1968–1972</i>							
Wagging Tail	9p	3	0.796 [0.664, 0.885]	1	0.776 [0.641, 0.870]	−0.020†	1.42
Env Test	18p	5	0.880 [0.762, 0.944]	4	0.920 [0.812, 0.968]	+0.040†	0.00
Dynamic Response	41p	3	0.580 [0.442, 0.706]	2	0.700 [0.562, 0.809]	+0.120	1.19
Natural Vibration	42p	4	0.660 [0.517, 0.778]	5	0.489 [0.353, 0.628]	−0.170†	6.33
Thermal Analysis	66p	5	0.680 [0.542, 0.792]	2	0.840 [0.715, 0.917]	+0.160	0.36
Heat Pipes	104p	14	0.720 [0.583, 0.825]	11 [med 12, {11, 12, 17}]	0.780 [0.648, 0.872]	+0.060	2.80
SP-5100 Shock	176p	47 [med 46.5]	0.521 [0.383, 0.655]	45 [med 45, {25, 45, 59}]	0.604 [0.463, 0.730]	+0.083	4.00
<i>Paper, arXiv 2026 (post-cutoff)</i>							
arXiv Fairness AI	53p	14	0.500 [0.366, 0.634]	15	0.400 [0.276, 0.538]	−0.100†	0.03
arXiv Megagauss	75p	36 [{36, 36}]	0.660 [0.522, 0.776]	32 [med 30, {28, 32}]	0.680 [0.542, 0.792]	+0.020†	9.46
arXiv Perovskite	80p	2	0.900 [0.786, 0.957]	3	0.740 [0.604, 0.841]	−0.160†	0.15

citation-heavy NASA survey for Mode B replication. Gemini C_0 median 46h/0.52 cov $\rightarrow$ C_1 median 45h/0.60 cov (range {25, 45, 59} across three judge re-runs of the byte-identical review, i.e. judge variance, not Pass-2 variance). Claude C_1 median 85h/0.81 cov (range {47, 85, 85, 97} across four judge re-runs, where the originally reported 47h was a low-end outlier). MIMO C_1 60h/0.23 cov. Claude C_0 not realizable: 176 page-images exceed Anthropic’s per-request payload limit even with the 1M-context beta. Spot-checking Claude C_1 shows the same bibliographic-expansion signature as Heat Pipes.

Modern arXiv reviews. Gemini Fairness AI: 14h/0.50 $\rightarrow$ 15h/0.40 (−10pp coverage, Mode A). Claude C_1 23h/0.70, and MIMO C_1 18h/0.32. Gemini Megagauss: C_0 median 36h/0.66 $\rightarrow$ C_1 median 30h/0.68 (high baseline-halluc physics review). Claude C_1 17h/0.92, and MIMO C_1 19h/0.26. Gemini Perovskite: 2h/0.90 $\rightarrow$ 3h/0.74 at $n=1$ (−6pp under multi-seed, near-noise-floor). Claude C_1 0h/0.90, and MIMO C_1 2h/0.90. The arXiv pattern (Mode A on 2/3) is opposite-direction from the NASA pattern despite comparable page counts. We attribute this to (a) arXiv full-color 200 DPI rendering carrying more visual signal per page, and (b) higher typesetting density forcing more aggressive C_1 compression at the same output budget.

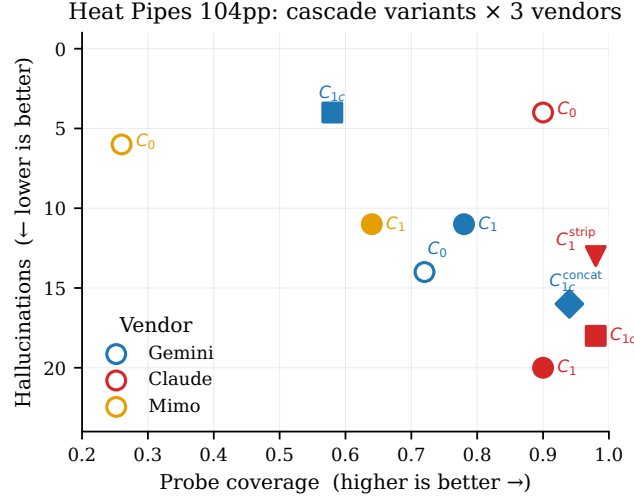


Figure 14. Heat Pipes 104pp Pareto frontier across vendors and cascade variants. Gemini cascade is small-Pareto-positive; Claude C_1 regresses on faithfulness (4h→20h), leaving C_0 on the Pareto frontier (the Mode-B signature); Mimo trades +5 halluc for +38pp coverage (Modes A+B compounded).

Table 7. Complete within-Claude C_0 -vs- C_1 comparison on the six NASA papers where Claude Opus 4.7’s multimodal C_0 fits the request payload (the data behind Figure 9b). Halluc. is the GPT-5.4 unsupported-claim count; cov. is probe-coverage on the reused Gemini reference/probes. Cascade reduces hallucinations on 5/6 (all but the Heat Pipes Mode-B cell); coverage gain tracks C_0 headroom (flat where C_0 is already near-ceiling, a clean Pareto win on Thermal where C_0 has room). Cells sorted by length. The Heat Pipes C_1 count is a single-seed snapshot (20h); its $n=5$ Pass-2 mean is 13.2h (Appendix K), so the +16 is an upper-end draw, but the 5/6 faithfulness direction holds under either.

Cell	len	C_0 h/cov	C_1 h/cov	Δ_h	Δ_{cov}
Wagging Tail	9p	5 / 0.959	2 / 0.980	−3	+0.021
Env Test	18p	3 / 1.000	2 / 1.000	−1	±0.000
Dynamic Response	41p	2 / 1.000	1 / 0.960	−1	−0.040
Natural Vibration	42p	5 / 0.915	4 / 0.851	−1	−0.064
Thermal Analysis	66p	11 / 0.920	4 / 1.000	−7	+0.080
Heat Pipes	104p	4 / 0.900	20 / 0.900	+16	±0.000
<i>direction (noise floor $\Delta_h \geq 1, \Delta_{cov} \geq 0.04$)</i>				5/6 ↓	—

E Mixed-model audio cascade: decomposition unlocks text-only models

The same-weights audio cascade is Gemini-only by API constraint (§8). In production one would not use one model for everything; one would use the best model per stage. This probe does exactly that, and is therefore *off-protocol*: it mixes vendors across stages, so it speaks to the generality of *decomposition*, not the same-weights claim. We take an *independent*, non-Google Pass-1 transcript from OpenAI gpt-audio (chunked, full quality—distinct from the Gemini reference used for judging), then run a text-only synthesis Pass-2 with three frontier text models, two of which (Claude Opus 4.7, GPT-5.4) cannot ingest audio at all on their own. Reviews are scored on the shared Pro reference and probes with the GPT-5.4 judge, plus a Claude-judge cross-check on the GPT-5.4-authored review to rule out self-grading.

The result is unambiguous (Table 8; Figure 9d): on all three cells, every mixed pipeline

beats Gemini’s *end-to-end* C_0 on both axes—coverage rises to 0.90–1.00 (vs C_0 0.63–0.88) and hallucinations fall to 2–9 (vs 11–17). Decomposition thus *unlocks* text-only frontier models for audio review and lets them outperform the native audio model’s end-to-end pass, using an off-the-shelf transcript. The GPT-5.4 self-grading concern is unfounded: the Claude judge scores the GPT-5.4-authored reviews *lower* on hallucinations (1/3/3 vs the GPT-5.4 judge’s 3/6/9), consistent with the $0.71\times$ inter-judge ratio (§6). The mixed pipeline trails the Gemini same-weights C_1 only where gpt-audio’s transcript is lossier than Gemini’s own ASR (MIT coverage 0.90 vs 0.98), i.e. the ceiling is set by Pass-1 transcript quality, exactly as the perceive–externalize–synthesize account predicts.

Table 8. Mixed-model audio cascade (off-protocol): independent gpt-audio Pass-1 transcript $\rightarrow$ text-only Pass-2 by each model, vs Gemini 3.1 Pro end-to-end C_0 . Hallucinations h / probe-coverage, GPT-5.4 judge; the parenthetical on GPT-5.4 is the Claude-judge cross-check. Claude and GPT-5.4 take no audio input, so their only route to an audio review is via the cascade—and it beats Gemini’s C_0 on every cell.

Cell	Gemini C_0 (e2e)	gpt-audio $\rightarrow$ Claude	gpt-audio $\rightarrow$ GPT-5.4	gpt-audio $\rightarrow$ Gemini
3B1B Attention	11h / 0.88	2h / 1.00	3h / 1.00 (1h / 1.00)	5h / 1.00
Karpathy	13h / 0.63	6h / 0.98	6h / 1.00 (3h / 1.00)	5h / 0.90
MIT 6.034	17h / 0.77	3h / 0.90	9h / 0.90 (3h / 0.92)	5h / 0.85

F Cross-model audio: Gemini 2.5 Flash (clean, no bitrate confound)

Table 9 reports a full-quality C_0 -vs- C_1 comparison with Gemini 2.5 Flash as the subject model (same reused Pro reference and probes, same GPT-5.4 judge), over five representative audio cells; Figure 9c plots it against the Pro arm. Unlike the Mimo audio arm, the audio is the full 32 kbps source (no OpenRouter 10MB cap), so there is no audio-quality confound. 2.5 Flash is a strong end-to-end audio reader: its C_0 coverage exceeds Gemini 3.1 Pro’s on four of five cells, leaving little headroom. The cascade direction then follows the inverse-baseline law—a clear win on the two headroom cells (MIT, Harvard), flat on the saturated NeurIPS panel, and a faithfulness regression on Karpathy (where C_0 is already 0.92/3h). Net: faithfulness improves on 3/5 and coverage on 1/5, a smaller and headroom-gated effect rather than the uniform audio win seen on Pro. Gemini 3 Flash was attempted as a second subject but its verbatim Pass-1 transcription returns empty under the model’s recitation filter on copyrighted audio, so the cascade is not realizable on it.

Table 9. Gemini 2.5 Flash audio C_0 -vs- C_1 (five cells, full-quality audio, GPT-5.4 judge on the reused Pro reference/probes). Sorted by length. The win tracks C_0 headroom: large where C_0 is weak (MIT, Harvard), absent where C_0 is already strong (NeurIPS, Karpathy).

Cell	min	C_0 h/cov	C_1 h/cov	Δ_h	Δ_{cov}
3B1B Attention	26	5 / 0.900	1 / 0.920	−4	+0.020
NeurIPS panel	31	5 / 0.980	6 / 0.959	+1	−0.020
Karpathy	42	3 / 0.922	9 / 0.902	+6	−0.020
MIT 6.034	47	7 / 0.833	4 / 0.958	−3	+0.125
Harvard Moot	82	19 / 0.600	12 / 0.620	−7	+0.020

G Iterative refinement ($C_{1\text{iter}}$): full results

The three-step pipeline and the high-level verdict are in §5; Figure 10b plots the per-cell movement and Table 10 gives the $n=5$ numbers.

Table 10. Iterative refinement ($C_{1\text{iter}}$) on 7 cells, $n=5$ seeds at $T=0.7$ (hallucinations: mean $\pm$ sd, coverage: mean). Δ columns are vs. that cell’s plain C_1 baseline, re-measured at the same $n=5$. “Pareto”: both axes improve beyond noise. “Trade”: faithfulness gain at a coverage cost. “Net-neg”: no faithfulness gain and a coverage loss. High-count Claude cells (SP-5100, Megagauss) carry large GPT-5.4 judge variance (§6). Baselines here are $n=5$ Pass-2 means and therefore differ from the single-seed Claude C_1 snapshots in Appendix D (e.g. Megagauss 40.6 vs the single-seed 17; SP-5100 mean 62.8 vs the judge-rerun median 85).

Cell (vendor)	C_1 baseline	$C_{1\text{iter}}$	Δ_{halluc}	Δ_{cov}	Verdict
Natural Vibration (Gemini)	7.8 $\pm$ 2.2 / 0.54	3.6$\pm$1.4 / 0.66	−4.2	+0.13	Pareto
Heat Pipes (Gemini)	14.4 $\pm$ 2.6 / 0.63	10.6 $\pm$ 3.1 / 0.46	−3.8	−0.17	trade
Heat Pipes (Claude)	14.8 $\pm$ 3.5 / 0.93	14.2 $\pm$ 4.4 / 0.74	−0.6	−0.20	net-neg
arXiv Fairness AI (Gemini)	15.6 $\pm$ 6.6 / 0.36	21.8 $\pm$ 1.0 / 0.36	+6.2	+0.01	net-neg
arXiv Fairness AI (Claude)	21.0 $\pm$ 3.9 / 0.58	21.8 $\pm$ 3.2 / 0.40	+0.8	−0.17	net-neg
SP-5100 (Claude)	62.8 $\pm$ 23 / 0.74	49.8 $\pm$ 13 / 0.56	−13.0	−0.18	trade
arXiv Megagauss (Claude)	40.6 $\pm$ 4.6 / 0.90	32.2 $\pm$ 8.5 / 0.64	−8.4	−0.25	trade

$C_{1\text{iter}}$ targets Mode B (a verbatim-quote requirement rules out unsupported expansions) but does not relieve Mode A: the rewrite step itself compresses, so coverage falls on every long cell, and the net verdict is the balance of recovered faithfulness against that compression cost. Two single-seed verdicts did not survive $n=5$: the Heat Pipes Claude Mode-B “fix” (now within noise) and an apparent “hallucinations increase” on SP-5100/Megagauss (both counts in fact *fall* at $n=5$; the single-seed rises were high-variance GPT-5.4 judge draws on 40–90-claim lists).

H Engineering interventions: chunked Pass-2 and citation stripping

Table 11. Heat Pipes Claude 104pp variants. Strip dominates plain C_1 at $n=1$ on both axes. Chunked combinations do not improve over strip alone. At $n=5$ multi-seed, the strip effect on C_1 hallucinations is statistically null ($\hat{\Delta} = -1.8 \pm 2.6$, 95% CI $[-6.9, +3.3]$).

Variant	Halluc.	Coverage	Words
C_0 (image baseline)	4	0.900	3,308
C_1 (single Pass-2)	20	0.900	2,077
C_1^{strip} (citations $\rightarrow$ [REF])	13	0.980	2,471
C_{1c} (chunked + merge)	18	0.980	2,769
C_{1c}^{concat} (chunked, no merge)	20	0.940	2,777
C_{1c}^{strip} (strip + chunked + merge)	18	0.980	2,673
$C_{1c}^{\text{strip+concat}}$ (strip + chunked, no merge)	16	0.980	2,766

Mode-B fix replication on 3 new long Claude cells. Table 12 reports the strip intervention across four Claude cells. On SP-5100, the published single-call snapshot C_1 47h $\rightarrow$ C_1^{strip} 24h

initially looked like a clean halving, but the unstripped C_1 review judged four times yields $\{47, 85, 85, 97\}$ (median 85), so the $-23h$ effect compares one strip call against a low-end outlier of the unstripped distribution. On Megagauss, strip has essentially no effect ($17 \rightarrow 19h$, $0.92 \rightarrow 0.94$ cov). On Fairness AI only 1 surface form matched our regexes, and the “stripped” Pass-2 produced a substantially different review ($23 \rightarrow 41h$, $0.70 \rightarrow 0.62$ cov), revealing that Claude’s text-only Pass-2 is itself non-deterministic. **Net Mode-B-fix status: zero replicated cells** (Heat Pipes retracted at $n=5$, Megagauss null, Fairness AI null-by-design, SP-5100 single-call-each).

Table 12. Mode B citation-strip intervention across four long Claude cells. “surface forms” = regex hits replaced with [REF].

Cell	len	surface forms	C_1 h	C_1 cov	C_1^{strip} h	C_1^{strip} cov
Heat Pipes (recall)	104p	55	20	0.900	13	0.980
SP-5100 (new)	176p	193	47	0.812	24	0.708
arXiv Megagauss (new)	75p	298	17	0.920	19	0.940
arXiv Fairness AI (new)	53p	1	23	0.700	41	0.620

Mode-A chunked Pass-2 replication and multi-seed verification. $n=1$ ($T=0$) results are consistent across four papers (Table 13): C_{1c}^{concat} (chunked, no merge) recovers coverage on long sources at a hallucination cost, while C_{1c} (chunked + merge) over-compresses: on Perovskite and Megagauss the merge variant collapses to 0.10 and 0.12 coverage respectively. At $n=5$ multi-seed (Table 14), the chunked-Pass-2 direction survives on Fairness AI and Perovskite but *not* on Megagauss, whose original $n=1$ (0.74cov) sits above the entire $n=5$ envelope $[0.64, 0.68]$ at $T=0.7$. We retract the original “chunked Pass-2 lifts coverage on all three arXiv counter-cells” claim.

Table 13. Chunked-Pass-2 ($n=1$, $T=0$) across four Gemini long-paper cells.

Cell	len	C_1 h/cov	C_{1c} h/cov	C_{1c}^{concat} h/cov	$\Delta_{\text{cov}}^{\text{concat}}$
Heat Pipes (recall)	104p	11 / 0.78	4 / 0.58	16 / 0.94	+0.16
arXiv Perovskite (new)	80p	3 / 0.74	1 / 0.10	13 / 0.82	+0.08
arXiv Megagauss (new)	75p	32 / 0.68	0 / 0.12	36 / 0.74	+0.06
arXiv Fairness AI (new)	53p	15 / 0.40	5 / 0.32	37 / 0.52	+0.12

Table 14. Multi-seed verification of C_{1c}^{concat} on the three arXiv cells ($n=5$ seeds, Gemini 3.1 Pro Pass-2, $T=0.7$).

Cell	C_1 baseline	orig $n=1$	$n=5$ halluc	$n=5$ cov	cov range	dir. survives
arXiv Fairness AI	15h / 0.40	37h / 0.52	31.6 ± 1.9	0.496 ± 0.055	$[0.440, 0.580]$	yes
arXiv Megagauss	32h / 0.68	36h / 0.74	42.2 ± 4.3	0.664 ± 0.017	$[0.640, 0.680]$	NO
arXiv Perovskite	3h / 0.74	13h / 0.82	16.6 ± 2.5	0.832 ± 0.033	$[0.800, 0.880]$	yes

Citation stripping on Gemini Natural Vibration ($n=5$ verified). At $n=1$ ($T=0$), strip moves halluc $5 \rightarrow 12$ but coverage $0.49 \rightarrow 0.81$ (+32pp). At $n=5$ ($T=0.7$, seeds 1–5): halluc

$\{15, 11, 9, 8, 8\}$ with mean 10.2, and coverage $\{0.660, 0.723, 0.702, 0.511, 0.766\}$ with mean 0.672. The strip-vs- C_1 direction (cov gain at halluc cost) survives all 5 seeds. The magnitude on coverage is +18pp (mean) versus the original +32pp single-seed reading.

I Multi-seed envelopes on the four headline cells

Figure 11b (main text) plots the envelopes; Table 15 gives the numbers.

Table 15. Per-cell multi-seed envelopes for the four headline cells (Gemini C_1 Pass-2 at $T=0.7$, seeds 1–5). “base-in-envelope?” checks whether the $n=1$ value sits within the per-seed range.

Cell	orig $n=1$	$n=5$ halluc	$n=5$ cov	base-in-envelope?
MIT 6.034 Winston	4h / 0.98	7.4 ± 1.5	0.971 ± 0.028	cov yes; halluc NO (low)
Sapolsky Behavioral	15h / 0.94	14.5 ± 2.5	0.898 ± 0.017	halluc yes; cov NO (high)
Heat Pipes Gemini	11h / 0.78	15.4 ± 2.7	0.732 ± 0.056	cov yes; halluc NO (low)
arXiv Perovskite	3h / 0.74	4.8 ± 2.2	0.844 ± 0.056	halluc yes; cov NO (low)

Two readings: (1) the published $T=0$ baselines are mostly representative of the $T=0.7$ distribution on the headline cells, since every cell’s $n=5$ envelope overlaps the published number on at least one axis. (2) Perovskite was the single seed-envelope point most far from the multi-seed mean (-0.10 pp coverage below it), and under multi-seed verification its cascade-cov loss reads -6 pp rather than -16 pp. We retain Perovskite as a near-noise-floor cell and treat its earlier “Mode-A counter-cell” label as overstated at $n=1$.

J E6 vision-model ladder and parametric load sweep

Table 16 reports the E6 logit-lens readout for both open VLMs across the load sweep $K \in \{1, 4, 8, 16, 24\}$ (facts packed into one stimulus, one queried). Answer accuracy is 1.00 for text and image at every K on both models, and the text–image readout-depth gap stays within ± 1 layer throughout (TOST-equivalent within a ± 1 -layer SESOI at every row). The Qwen2.5-VL-32B variant was excluded: it verbalizes the answer in a form our single-token matcher cannot capture (a spurious acc = 0 artifact), so its readout depths are unreliable, and the 3B and 7B models carry the claim.

K Three sources of stochasticity

Detailed decomposition. (a) *Claude Pass-2 generation variance.* Same input + same API + same prompt: Heat Pipes Claude C_1 swings $20 \rightarrow 9$ between two runs (strip $13 \rightarrow 14$ more stable), and the deltas are large enough that the strip effect cannot be distinguished from Pass-2 noise. Claude Opus 4.7 deprecates the temperature parameter so determinism cannot be enforced from the SDK. (b) *Gemini Pass-2 generation variance.* At temperature=0.0, two of three Gemini Pass-2 re-runs produced byte-identical output (MD5 match) on Heat Pipes / SP-5100, and the third differed in similar length (1929 vs 1815 words, $\Delta h = +6$). Gemini is therefore mostly but not strictly deterministic, consistent with GPU-level non-associativity. (c) *GPT-5.4 high-reasoning judge variance.* Essentially deterministic on low-count cells (± 1 on Heat Pipes Claude across three re-judgments) but substantially stochastic on high-count cells

Table 16. E6 cross-modal logit-lens across two open VLMs and a K -fact load sweep. *text/img depth* = first layer the answer token is top-1. *gap CI90* = 90% CI on (img–text) depth in layers. *equiv* = TOST-equivalent within ± 1 layer. Accuracy is 1.00 for both modalities in every row.

Model	K	text acc	img acc	text depth	img depth	gap CI90 (equiv)
Qwen2.5-VL-3B	1	1.00	1.00	32.00	32.00	[0.00, 0.00] (yes)
Qwen2.5-VL-3B	4	1.00	1.00	32.25	32.06	[−0.41, 0.04] (yes)
Qwen2.5-VL-3B	8	1.00	1.00	32.19	32.06	[−0.27, 0.02] (yes)
Qwen2.5-VL-3B	16	1.00	1.00	32.75	32.06	[−0.88, −0.49] (yes)
Qwen2.5-VL-3B	24	1.00	1.00	32.62	32.00	[−0.83, −0.42] (yes)
Qwen2.5-VL-7B	1	1.00	1.00	24.31	24.44	[−0.02, 0.27] (yes)
Qwen2.5-VL-7B	4	1.00	1.00	24.25	24.12	[−0.27, 0.02] (yes)
Qwen2.5-VL-7B	8	1.00	1.00	24.00	24.00	[−0.15, 0.15] (yes)
Qwen2.5-VL-7B	16	1.00	1.00	24.06	24.12	[−0.04, 0.17] (yes)
Qwen2.5-VL-7B	24	1.00	1.00	24.06	24.12	[−0.04, 0.17] (yes)

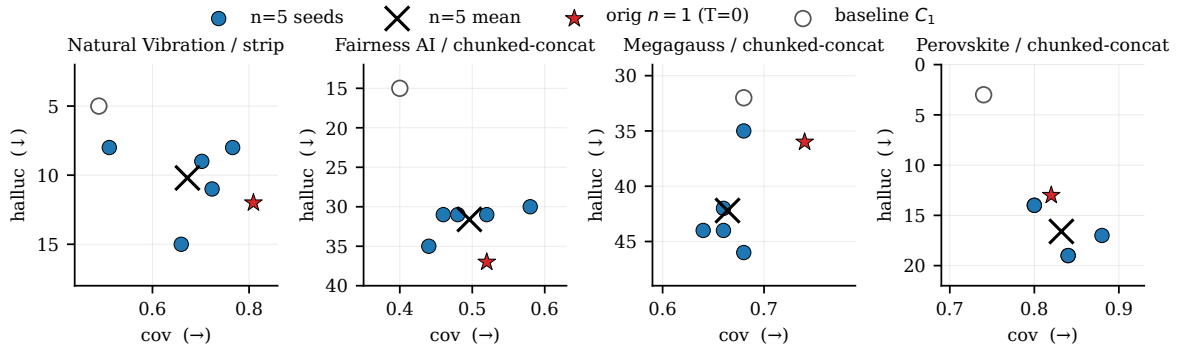


Figure 15. $n=5$ multi-seed envelopes for four intervention-verification cells. Blue dots: five Pass-2 seeds at $T=0.7$. Black $\times$: $n=5$ mean. Red star: originally-published $n=1$ at $T=0$. Open circle: cell’s C_1 baseline (no intervention). NV strip, Fairness AI, Perovskite: $n=1$ sits within $n=5$ envelope. Megagauss: $n=1$ is outside on both axes (lower halluc and higher cov than any of 5 seeds), so the published Megagauss chunked-Pass-2 cov-gain claim is retracted at $n=5$.

(± 25 on SP-5100, ± 17 on the byte-identical SP-5100 Gemini review). We attribute this to long unsupported-claim lists running up against our 24k-token completion budget, so the judge truncates and resamples across calls. Coverage is stable on every cell across judge re-runs.

$n=5$ multi-seed verification on Heat Pipes Claude. C_1 halluc {9, 11, 11, 15, 20} (mean 13.2, stdev 4.4, range 9–20), and C_1^{strip} halluc {5, 11, 13, 14, 14} (mean 11.4, stdev 3.8, range 5–14). Coverage means identical (0.936 both) with similar spread. Strip effect on hallucinations at $n=5$: $\hat{\Delta} = -1.8$, $\text{SE} \approx 2.6$, 95% CI [−6.9, +3.3], which crosses zero. **The Mode B fix is statistically null on Claude/Heat Pipes at $n=5$.**

L Inter-judge details

Figure 11a (main text) plots the 22 cell-condition pairs (21 Gemini cells plus the Thermal-Claude cross-vendor cell); the tables below give the per-cell counts on the original 10 cells

Table 17. Stochasticity decomposition. Top: Claude Pass-2 generation variance (different reviews, same prompt, judged once each). Bottom: GPT-5.4 judge variance on the *same* Claude C_1 review file (one review, judged multiple times).

Cell	Condition	Run 1	Run 2	Δh	Δcov
<i>Claude Pass-2 generation variance</i>					
Heat Pipes 104p	C_1	20h / 0.90	9h / 0.90	−11	±0.00
	C_1^{strip}	13h / 0.98	14h / 0.88	+1	−0.10
SP-5100 176p	C_1	47h / 0.81	79h / 0.81	+32	±0.00
	C_1^{strip}	24h / 0.71	97h / 0.77	+73	+0.06
Fairness AI 53p	C_1	23h / 0.70	27h / 0.66	+4	−0.04
	C_1^{strip}	41h / 0.62	17h / 0.60	−24	−0.02
<i>GPT-5.4 judge variance (same review, judged multiple times)</i>					
Heat Pipes 104p	C_1 (calls 1–3)	20h / 0.90	21h / 0.92	+1	+0.02
SP-5100 176p	C_1 (calls 1–4)	47h / 0.81	97h / 0.81	+50	±0.00

(per-cell deltas for the remaining cells are released in the repository).

Table 18. Inter-judge hallucination counts on the original 10 cells. Direction ✓ means both judges agree on the sign of $C_0 \rightarrow C_1$. The full 21-cell direction counts are 19/21 coverage and 14/21 hallucination (§6).

Cell	GPT-5.4		Claude		Direction agree?
	C_0	C_1	C_0	C_1	
Karpathy review	13	3	7	4	✓
Karpathy mtg-min	4	2	1	0	✓
SNAC mtg-min	14	11	10	11	×
Wagging Tail (9pp)	3	1	2	2	×
Env Test (18pp)	5	4	0	2	×
Dynamic Response (41pp)	3	2	1	0	✓
Natural Vibration (42pp)	4	5	2	4	✓
Thermal Analysis (66pp)	5	2	2	1	✓
Heat Pipes (104pp)	14	11	12	11	✓
Thermal Analysis Claude	11	4	9	4	✓

Per-probe agreement (the fraction of probes the two judges classify identically as COVERED / MISSING) ranges 0.882–1.000 across the (cell, condition) pairs, median ≈ 0.94 . Two of the three hallucination-direction disagreements on the original 10 cells (Wagging Tail, Env Test) are near-tie cells with ≤ 5 hallucinations under both judges. The third (SNAC) is driven by the judges assigning different C_0 baselines (14 vs 10) while agreeing on C_1 .

M Probe-circularity and reference-bias check tables

Figure 11c,d (main text) plots both checks; the tables below give the numbers.

The Thermal hallucination-direction flip under EasyOCR is a reference-side artifact: EasyOCR’s transcript is $\approx 21\%$ shorter and loses figure captions, so claims the cascade correctly reproduces read as unsupported against the noisier reference.

Table 19. Inter-judge probe-coverage counts (probes covered out of n) on the original 10 cells.

Cell	n	GPT-5.4		Claude		Direction agree?
		C_0	C_1	C_0	C_1	
Karpathy review	51	32	49	38	49	✓
Karpathy mtg-min	50	8	17	13	21	✓
SNAC mtg-min	49	14	26	15	28	✓
Wagging Tail (9pp)	49	39	38	40	38	✓
Env Test (18pp)	50	44	46	46	49	✓
Dynamic Response (41pp)	50	29	35	32	36	✓
Natural Vibration (42pp)	47	31	23	32	28	✓
Thermal Analysis (66pp)	50	34	42	34	45	✓
Heat Pipes (104pp)	50	36	39	39	41	✓
Thermal Analysis Claude	50	46	50	46	49	✓

Table 20. Probe-circularity check on 7 cells spanning the cascade-direction spectrum. “Original” = published numbers (GPT-5.4-extracted probes, GPT-5.4 judge); the other columns re-extract probes via Claude and score under each judge.

Cell (orig → Claude-extracted probes)	Δ_{cov} (orig)	Claude-pr. / GPT	Claude-pr. / Claude
Karpathy review (51 → 55) [audio]	+0.333	+0.400	+0.236
MIT 6.034 (48 → 50) [audio]	+0.208	+0.180	+0.120
Heat Pipes 104pp (50 → 60) [paper]	+0.060	+0.050	+0.033
Natural Vibration 42pp (47 → 50) [paper]	−0.170	−0.160	−0.200
arXiv Fairness AI 53pp (50 → 50) [paper]	−0.100	−0.100	+0.040
arXiv Megagauss 75pp (50 → 60) [paper]	+0.020	+0.000	+0.000
arXiv Perovskite 80pp (50 → 54) [paper]	−0.160	−0.167	−0.111

Table 21. Reference-bias check on 4 cells. The non-Gemini reference rows replace OCR/ASR only, and probes are re-extracted by Claude. “orig” = Gemini reference + GPT-5.4 probes + GPT-5.4 judge.

Cell	Reference	Δ_h orig	Δ_h alt	Δ_{cov} orig	Δ_{cov} alt
3B1B Attention 26m	Whisper-base / GPT-5.4	−11	−7	+0.080	+0.120
3B1B Attention 26m	Whisper-base / Claude	−11	−6	+0.080	+0.060
Karpathy review 42m	Whisper-base / GPT-5.4	−10	−4	+0.333	+0.353
Karpathy review 42m	Whisper-base / Claude	−10	−6	+0.333	+0.176
Wagging Tail 9p	EasyOCR / GPT-5.4	−2	−1	−0.020	−0.019
Wagging Tail 9p	EasyOCR / Claude	−2	−3	−0.020	−0.038
Thermal Analysis 66p	EasyOCR / GPT-5.4	−3	+2	+0.160	+0.094
Thermal Analysis 66p	EasyOCR / Claude	−3	+1	+0.160	+0.094

N Per-cell hallucinations per 1k output words

End-to-end hallucination rates differ per vendor (3.7 / 4.2 / 4.9 per 1k), and cascade brings Gemini and Claude to a common ≈ 1.4 per 1k floor on the moderate-length paper. Mimo does not converge to this floor.

Claude’s per-1k hallucination rate jumps from 1.2 (C_0) to 9.6 (C_1) on Heat Pipes, an $8\times$ increase that survives length normalization. The same model on the same protocol on Thermal 66pp moved $3.7 \rightarrow 1.4$. Citation stripping pulls Heat Pipes Claude back to 5.3/1k. Gemini

Table 22. Hallucination density (per 1k output words) on Thermal Analysis 66pp.

	C_0	C_1
Gemini 3.1 Pro	4.2	1.4
Claude Opus 4.7	3.7	1.4
Mimo v2-omni	4.9	5.4

Table 23. Hallucination density (per 1k output words) on Heat Pipes 104pp. Claude C_1 at 9.6/1k is the largest value in our dataset and dwarfs Claude’s Thermal C_1 rate of 1.4/1k, supporting the Mode B interpretation as a length- and citation-density-triggered effect rather than a per-token verbosity effect.

	C_0	C_1
Gemini 3.1 Pro	7.6	6.1
Claude Opus 4.7	1.2	9.6
Claude Opus 4.7 (C_1^{strip})	—	5.3
Mimo v2-omni	7.0	6.9

and Mimo on the same paper sit at 6–8/1k regardless of cascade variant, indicating that the 1.4/1k floor reached on Thermal is not a property of the cascade in general, but of the specific (model, source) interaction.

O Specific hallucination examples (Heat Pipes Claude)

The 20 unsupported claims from Claude C_1 on Heat Pipes 104pp are dominated by training-prior expansions of bibliographic abbreviations:

- “Cotter’s 1965 analysis (LA-3246-MS)”: the transcript cites only “Reference 2 (Cotter)”, so the year and report number are training-data details.
- “correlation from Allingham and McEntire (1961)”: the transcript only says “Ref. 12”, so the author names and year are training-data details.
- “McAdams’ laminar horizontal-tube formula”: the transcript says “Reference 13 for laminar condensation on horizontal tubes”, so “McAdams” is a training-data detail.
- “Vapor chamber-fin radiator (Haller, Lieblein, and Lindow at NASA Lewis)”: the transcript says only “proposed by Haller, et al of NASA”.

One outright misreading: the transcript states the no-boiling condition as $p_v - p_\ell \leq 2\sigma \cos \theta / r_c$, but Claude’s review writes $\geq$. This is a sign error in Pass-2 reading of Pass-1 text, not training-prior leak.

P Code and data availability

All code, prompts, per-cell judgments, and the mechanistic-experiment artifacts (including the E8/E9 substrate probes, the running-example run directory, and the within-Claude cross-vendor comparison) are released in the project repository, whose README documents the pipelines and the exact commands to reproduce every cell, table, and figure in this paper.